A Practical record on

EXPLORATORY DATA ANALYSIS USING EXCEL (BCADA1P2)

Submitted in partial fulfillment of the requirement for

Bachelor of Computer Applications in Data Analytics

Submitted by

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ST. JOSEPH'S UNIVERSITY, BENGALURU -27

DEPARTMENT OF ADVANCED COMPUTING

2023-2024

CERTIFICATE

This is to certify Mr./Mrs.

“……

……

…” has successfully completed in 1st semester Data analytics lab using Excel prescribed by ST. JOSEPH’S UNIVERSITY, under Department of Advanced Computing - BCA Data Analytics course in the year 2023 - 2024.

Signature of the Teacher in-charge of the Batch

Head of the Department

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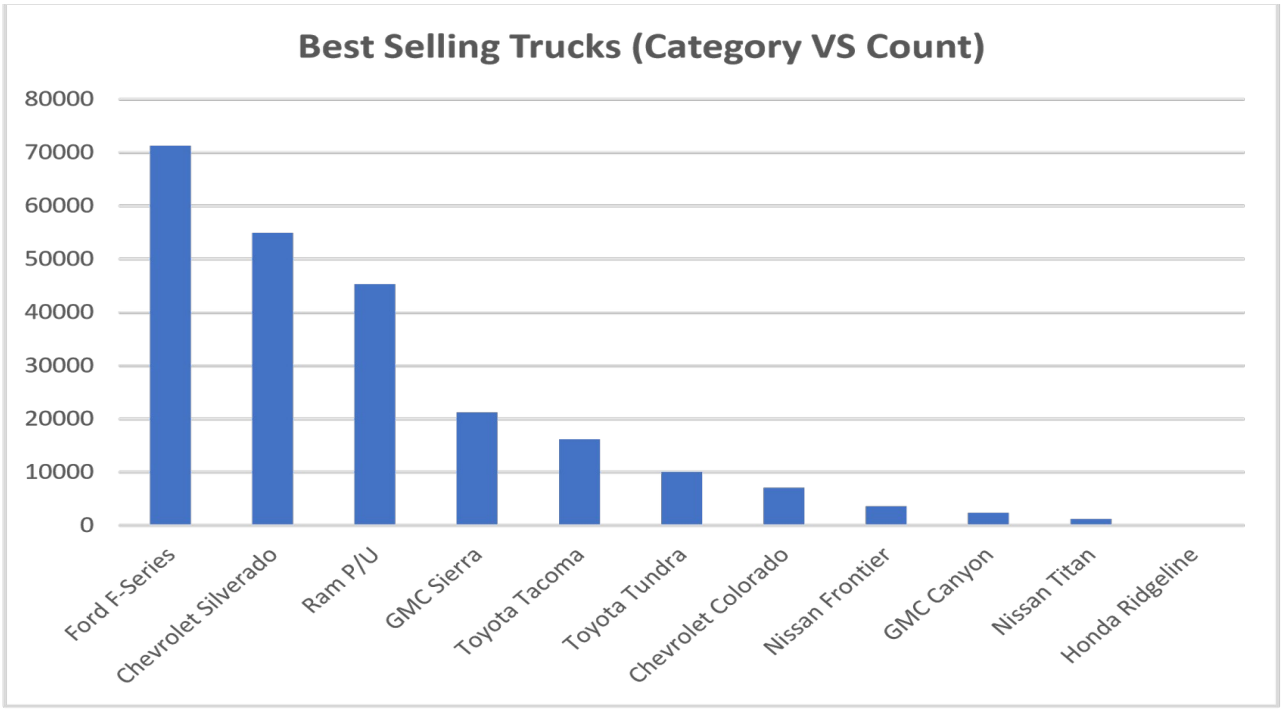
ASSIGNMENT – 1

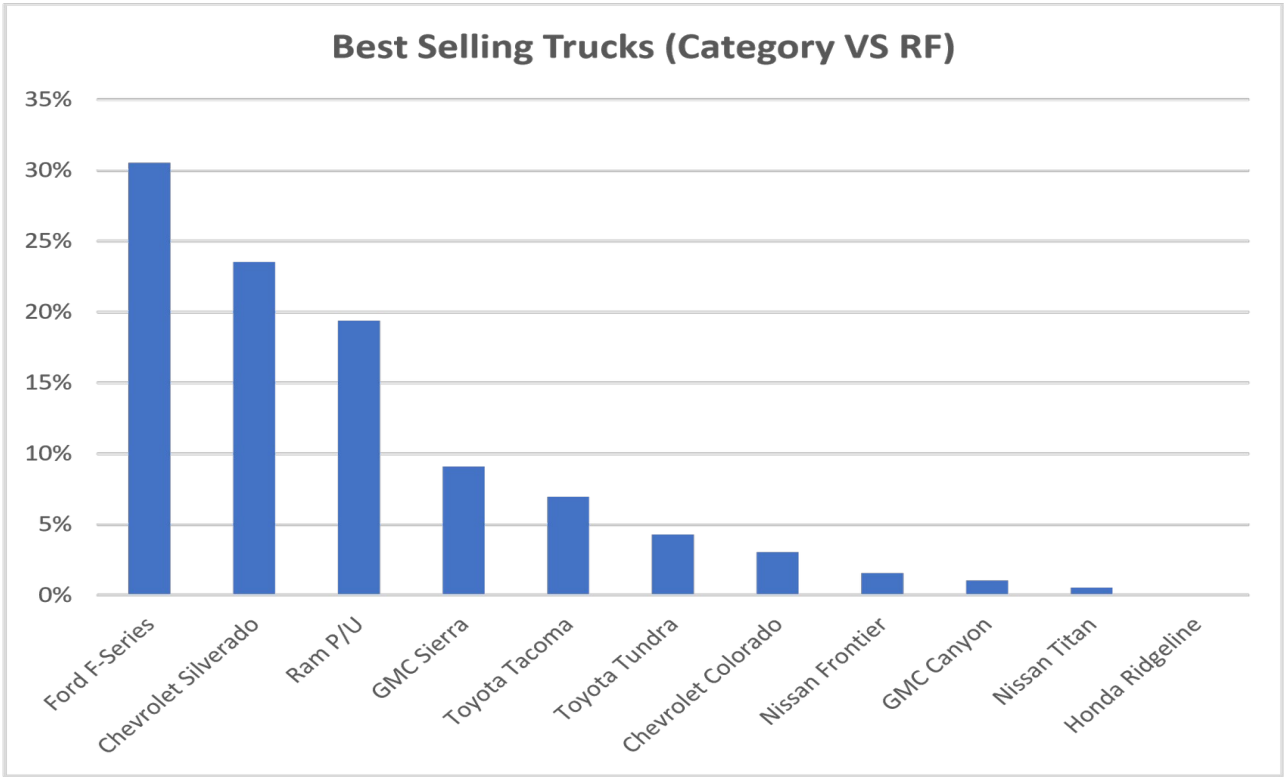
GIVEN DATA

Truck Model Sold
Toyota Tacoma
Ram P/U
Chevrolet Silverado
Chevrolet Silverado
Toyota Tundra
GMC Sierra
Ram P/U
Toyota Tacoma
Ford F-Series
Ram P/U
Ford F-Series
Chevrolet Silverado
Chevrolet Silverado
Ford F-Series
Ram P/U

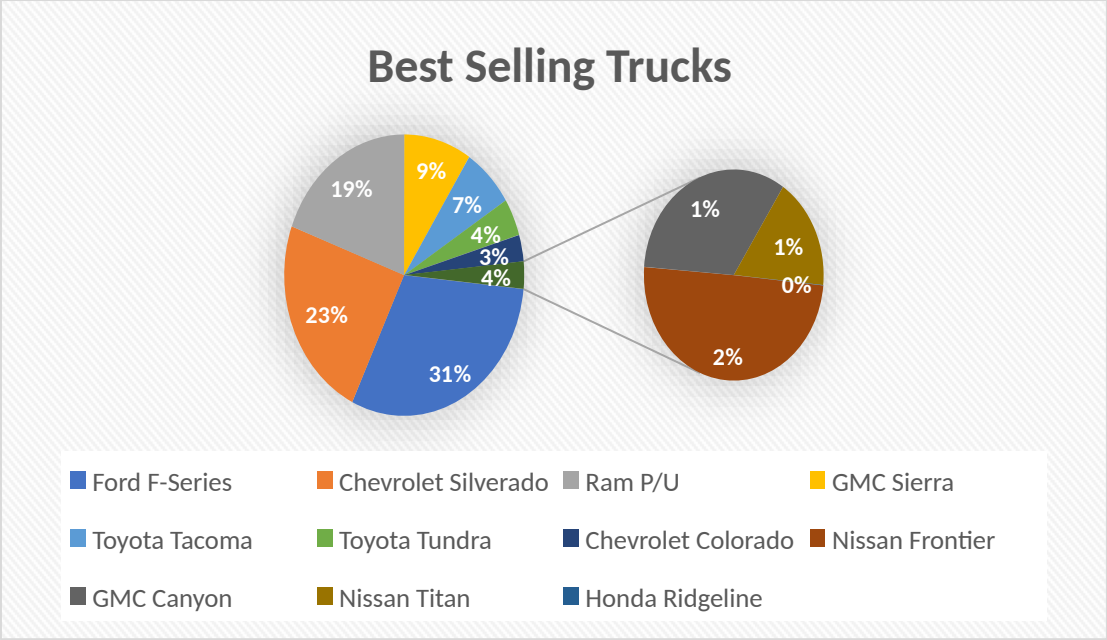
1. **Find the frequency of each category and calculate the relative frequency. Create a bar plot Category vs Count and Category vs Relative Frequency.**

Truck Model Sold	Count	Relative Frequency
Ford F-Series	71332	31%
Chevrolet Silverado	54977	24%
Ram P/U	45310	19%
GMC Sierra	21241	9%
Toyota Tacoma	16230	7%
Toyota Tundra	10057	4%
Chevrolet Colorado	7114	3%
Nissan Frontier	3645	2%
GMC Canyon	2423	1%
Nissan Titan	1268	1%
Honda Ridgeline	4	0%
Grand Total	**233601**	**100%**


Best Selling Trucks (Category VS Count)
80000
70000
60000
50000
40000
30000
20000
10000
0
Ford F-Series
Chevrolet Silverado
Ram P/U
GMC Sierra
Toyota Tacoma
Toyota Tundra
Chevrolet Colorado
Nissan Frontier
GMC Canyon
Nissan Titan
Honda Ridgeline


Best Selling Trucks (Category VS RF)
35%
30%
25%
20%
15%
10%
5%
0%
Ford F-Series
Chevrolet Silverado
Ram P/U
GMC Sierra
Toyota Tacoma
Toyota Tundra
Chevrolet Colorado
Nissan Frontier
GMC Canyon
Nissan Titan
Honda Ridgeline

2. **Similarly plot a pie chart.**



3. **Plotting table using Pivot Table.**

To plot Pivot table in Excel the user needs to follow the following steps.

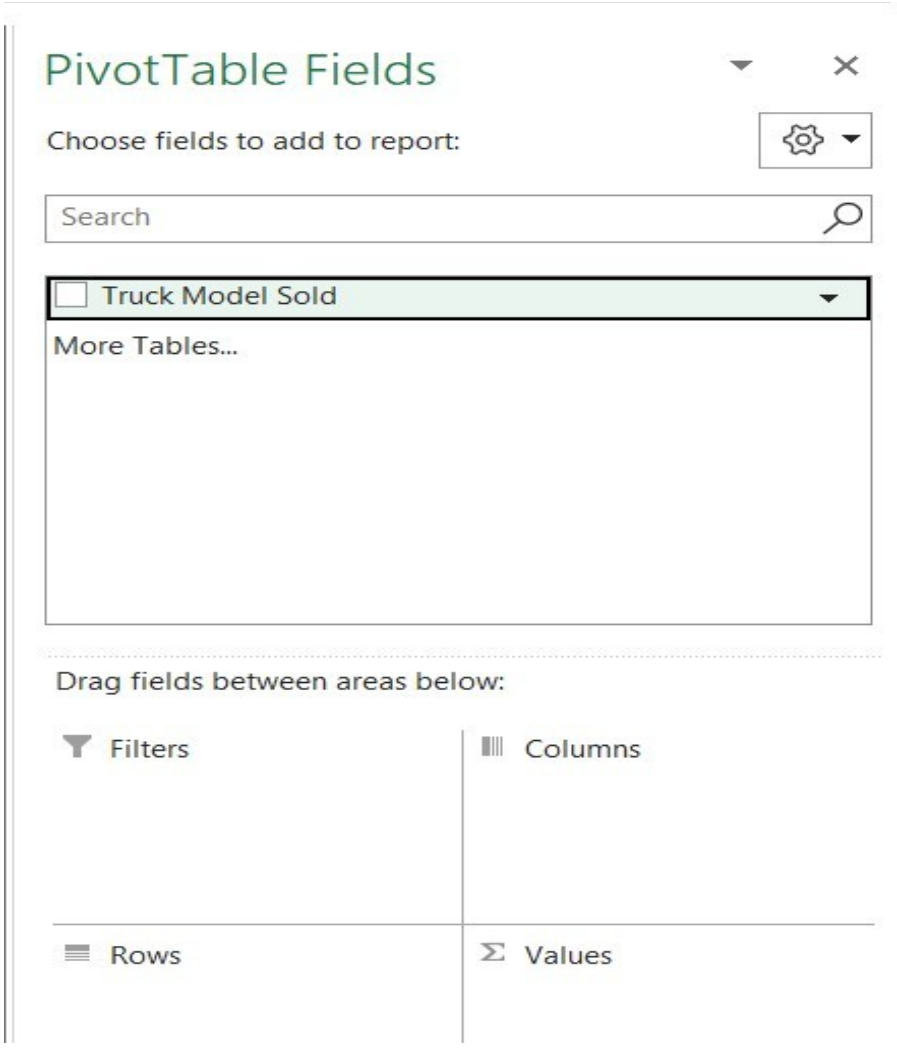
Step – 1 : ***Select the given data.***

	A	B
1	**Truck Model Sold**	
2	Toyota Tacoma	
3	Ram P/U	
4	Chevrolet Silverado	
5	Chevrolet Silverado	
6	Toyota Tundra	
7	GMC Sierra	
8	Ram P/U	
9	Toyota Tacoma	
10	Ford F-Series	
11	Ram P/U	
12	Ford F-Series	
13	Chevrolet Silverado	
14	Chevrolet Silverado	
15	Ford F-Series	
16	Ram P/U	

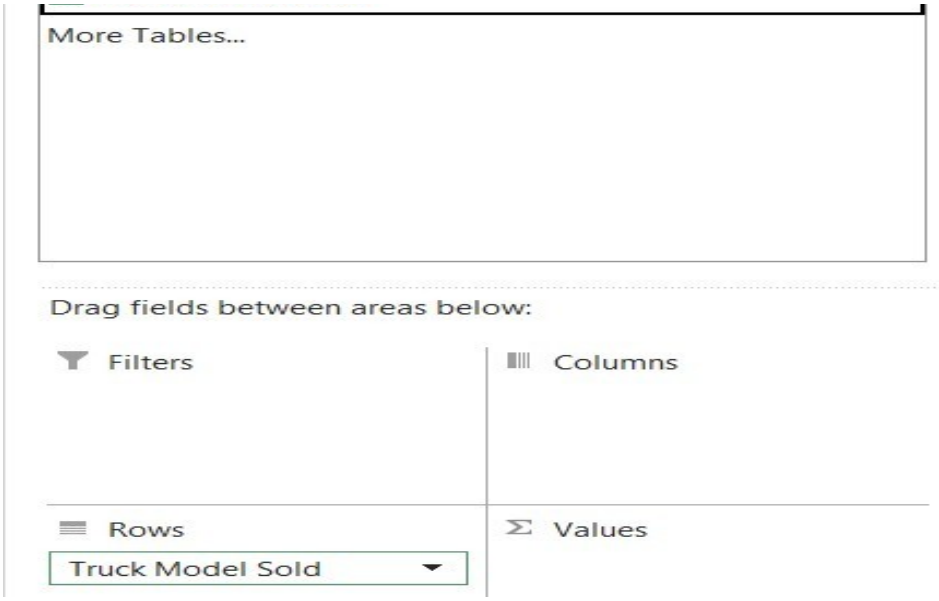
Step – 2 ***: After selecting the given data go to Insert Tab.***



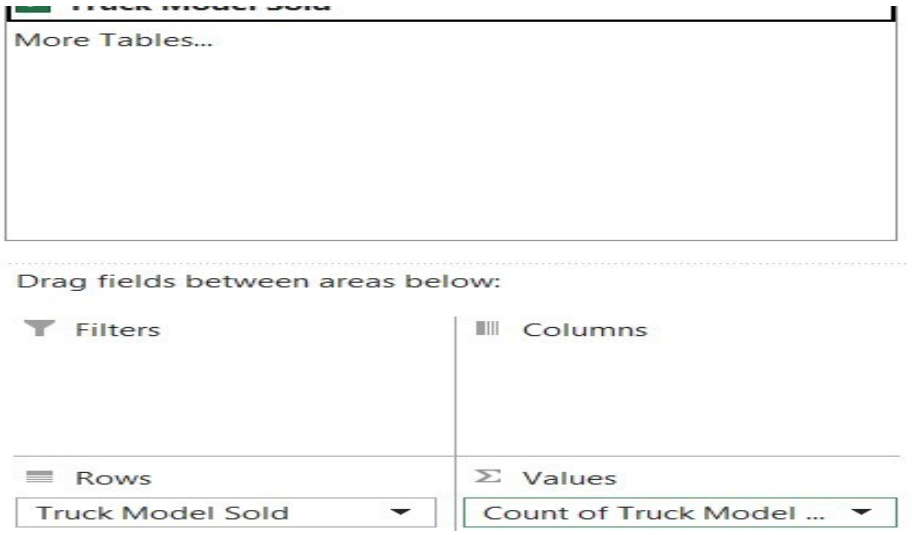
Step – 3 : ***Select 'Truck Models Sold'.***


PivotTable Fields
Choose fields to add to report:
Search
Truck Model Sold
More Tables...
Drag fields between areas below:
Filters
Columns
Rows
Values

Step – 4:Drag‘Truck Models Sold’ to ‘Rows Column’.



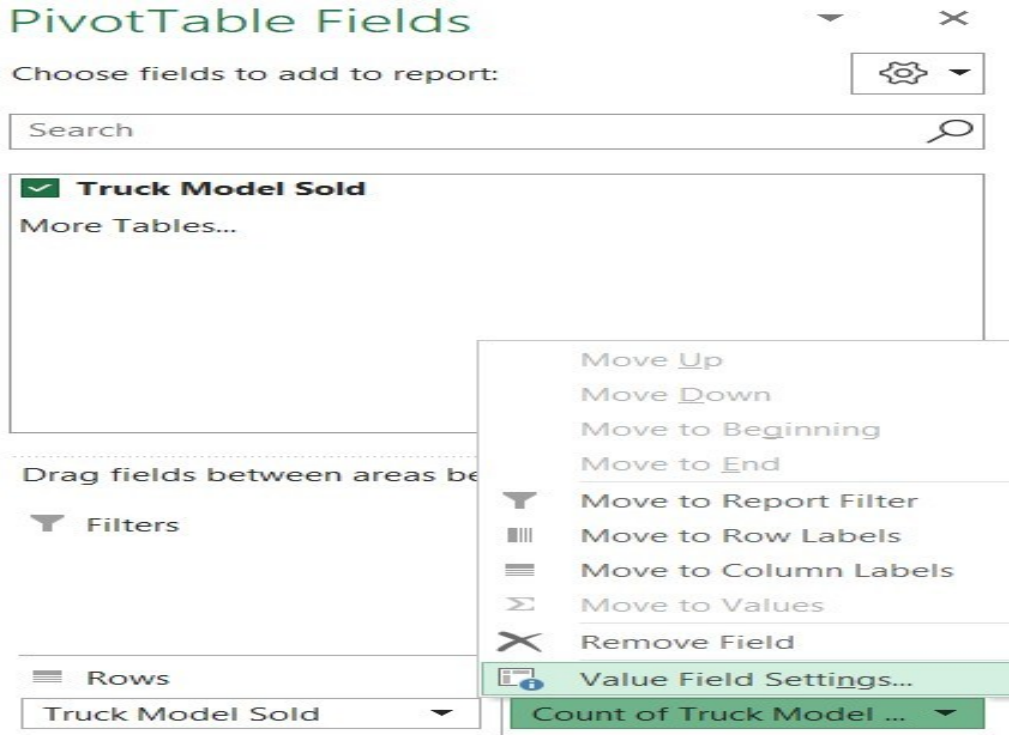
Step – 5: Drag ‘Truck Models Sold’ to ‘Values Column’.



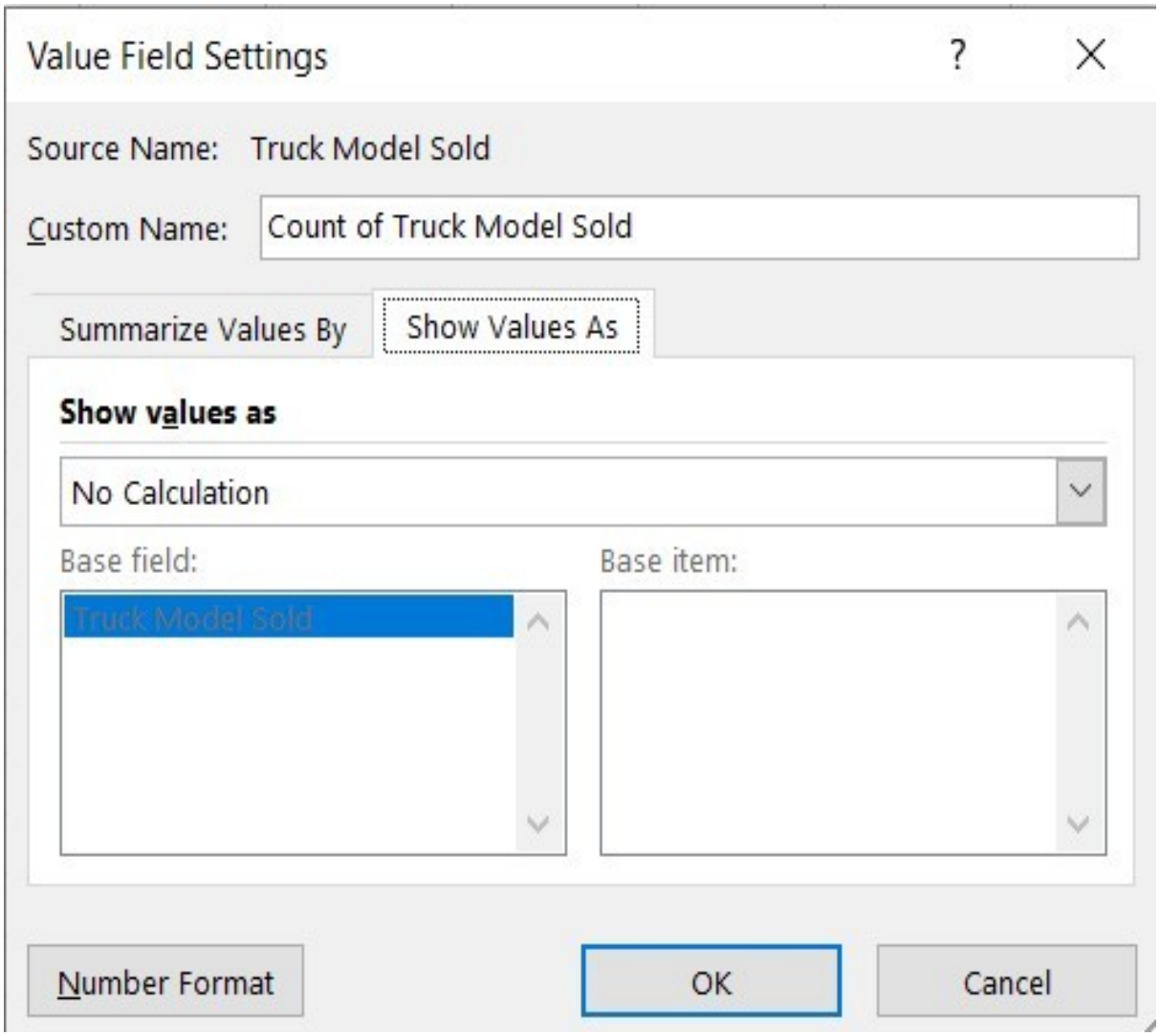
Step – 6 : Generated a pivot table.

Row Labels	Count of Truck Model Sold
Chevrolet Colorado	7114
Chevrolet Silverado	54977
Ford F-Series	71332
GMC Canyon	2423
GMC Sierra	21241
Honda Ridgeline	4
Nissan Frontier	3645
Nissan Titan	1268
Ram P/U	45310
Toyota Tacoma	16230
Toyota Tundra	10057
(blank)	
Grand Total	**233601**

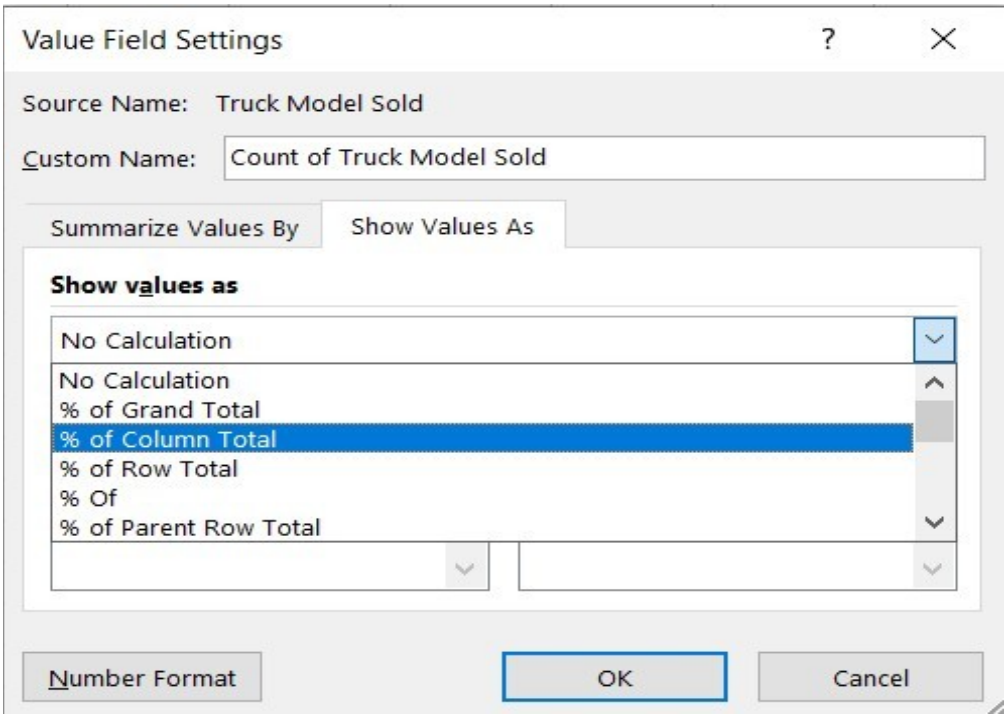
<u>Step – 7:</u> To get Relative Frequency of the data, select "Value Field Settings"



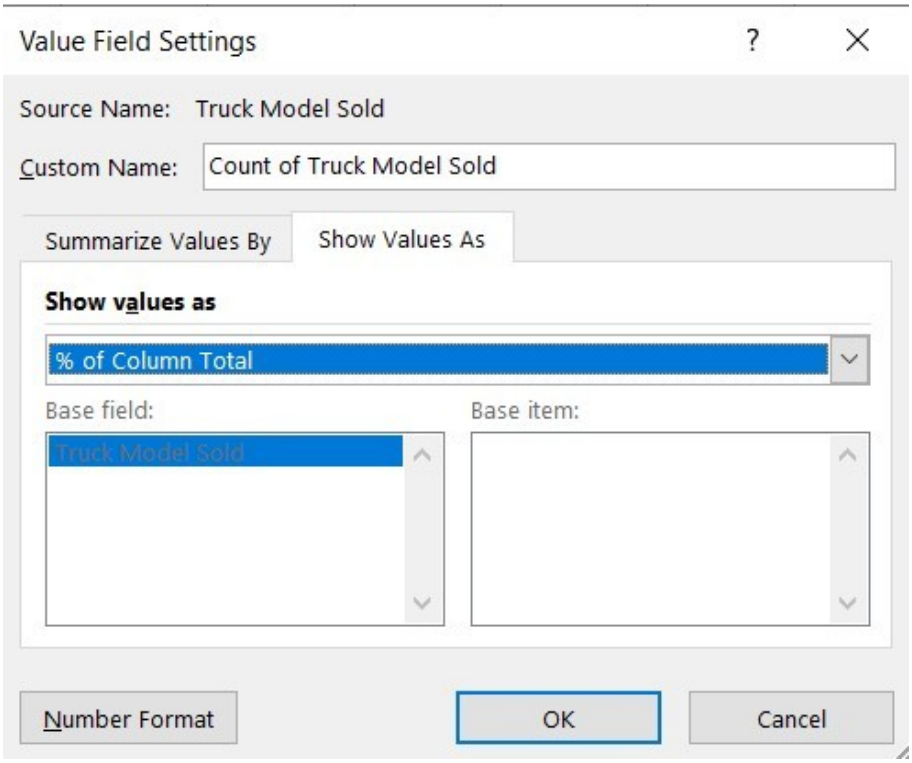
Step – 8: ***Select “Show Values As” .***



Step – 9 : Select "% of Column Total" option and select OK.



Step – 10 : Select OK.

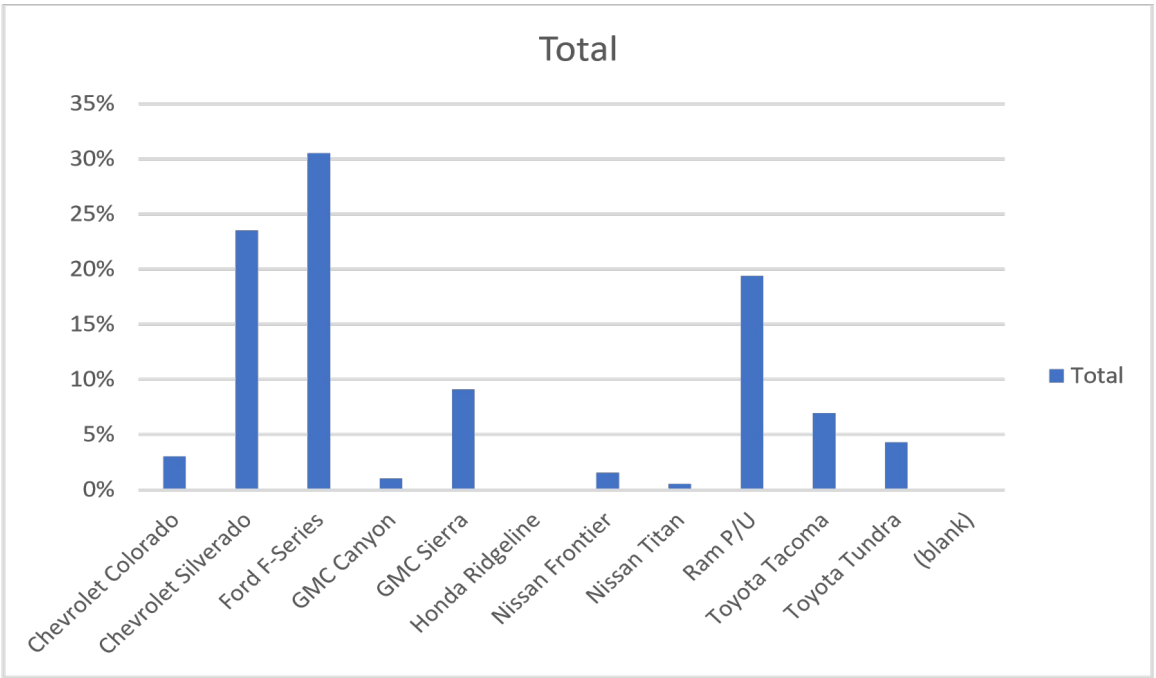


Step – 11 : FINAL OUTPUT OF PIVOT TABLE.

Row Labels	Count of Truck Model Sold
Chevrolet Colorado	7114
Chevrolet Silverado	54977
Ford F-Series	71332
GMC Canyon	2423
GMC Sierra	21241
Honda Ridgeline	4
Nissan Frontier	3645
Nissan Titan	1268
Ram P/U	45310
Toyota Tacoma	16230
Toyota Tundra	10057
(blank)	
Grand Total	**233601**

Row Labels	Count of Truck Model Sold
Chevrolet Colorado	3.05%
Chevrolet Silverado	23.53%
Ford F-Series	30.54%
GMC Canyon	1.04%
GMC Sierra	9.09%
Honda Ridgeline	0.00%
Nissan Frontier	1.56%
Nissan Titan	0.54%
Ram P/U	19.40%
Toyota Tacoma	6.95%
Toyota Tundra	4.31%
(blank)	0.00%
Grand Total	**100.00%**

Step – 12 : Bar Graph.



Summary.

Here we can see that There is a sale of 233601 units trucks in a year. In this year, Ford F-Series truck model is the best-selling truck of the year with the sales of 71332 units occupying 31% of the trucks market. Whereas Honda Ridgeline had did the sales of only 4 units which is the least.

Here we have used =uniqueif() to find the unique record , =countif() for finding count and =count/total for relative frequency.

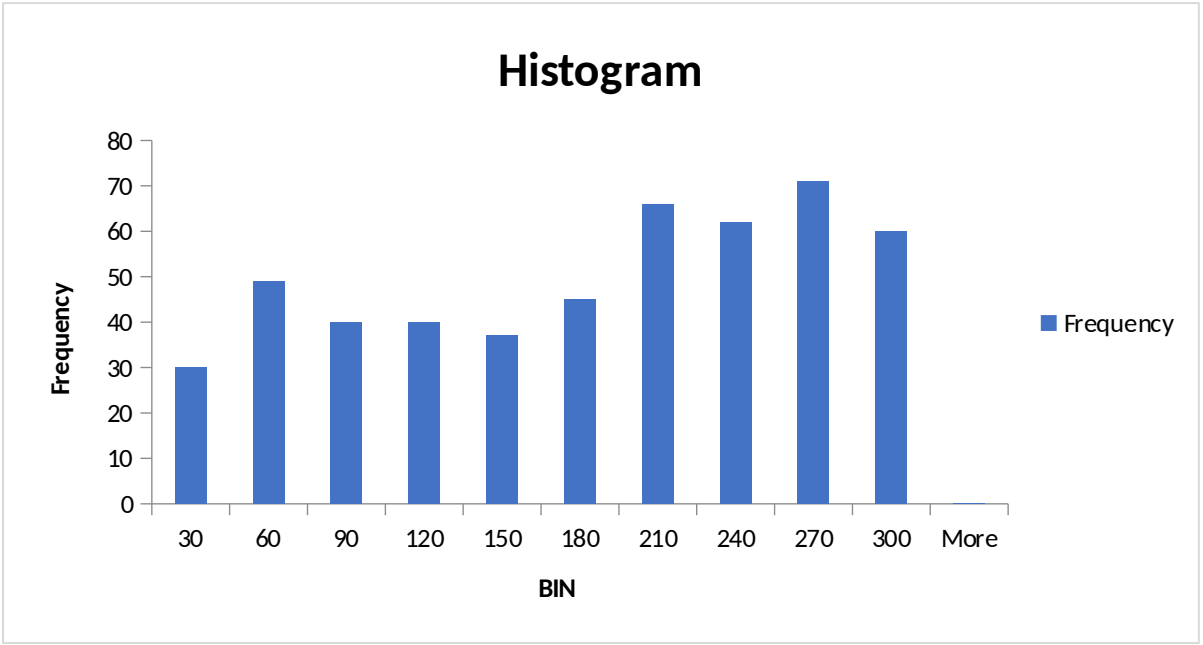
ASSIGNMENT – 2

GIVEN DATA

Customer Waiting Time (seconds)
51
95
68
243
167
288
108
83
48
251
189
128
60
290
259

1.Create a range of bins and find the frequency and relative frequency.

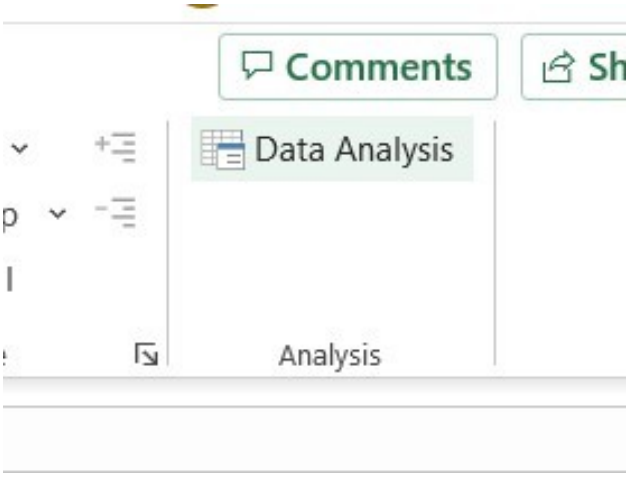
Bin Range	Frequency	Relative Frequency
30	30	6%
60	49	10%
90	40	8%
120	40	8%
150	37	7%
180	45	9%
210	66	13%
240	62	12%
270	71	14%
300	60	12%
Total Frequency	**500**	**100%**



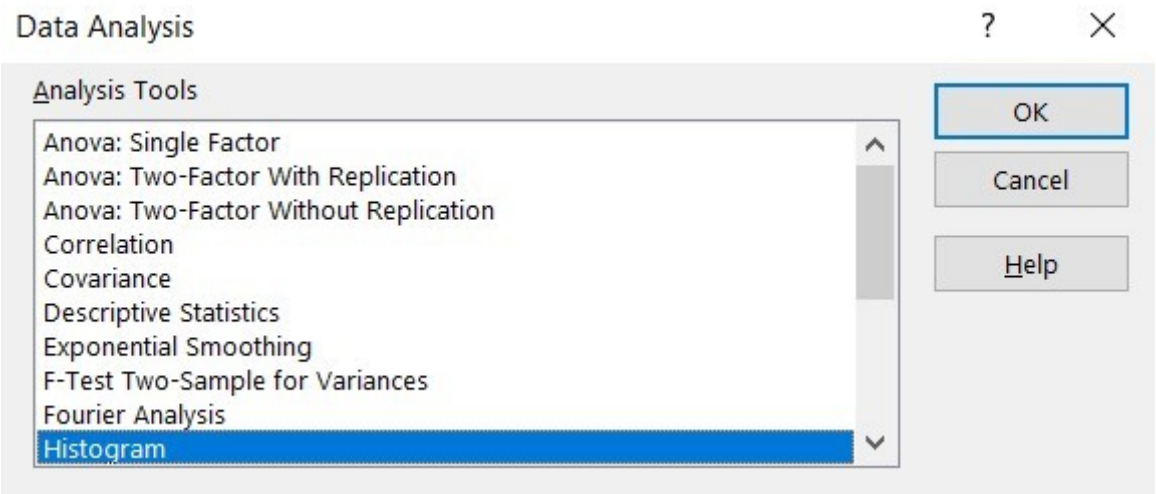
1. ***<u>Another way use data analysis toolpak to calculate the Frequency table and plot Histogram using 'Histogram' Function.</u>***

   To generate table using data analysis toolpak, the user needs to follow the steps. Such as:-

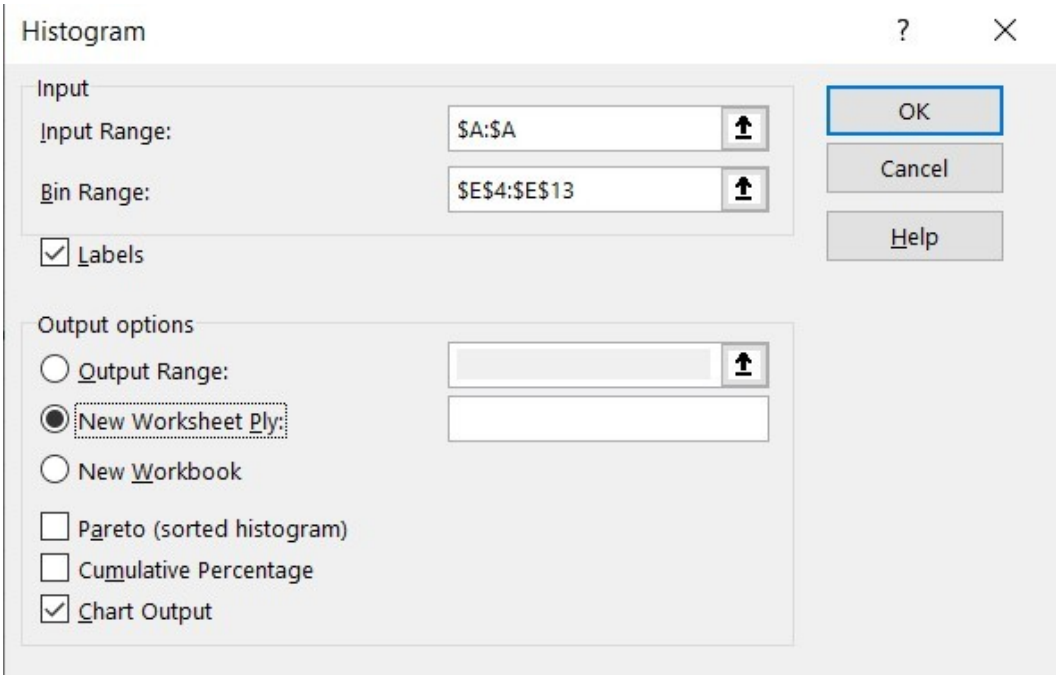
Step – 1 : Open 'Data' tab and select 'Data Analysis' from 'Analysis' section.



Step – 2 : Select 'Histogram' and click OK.

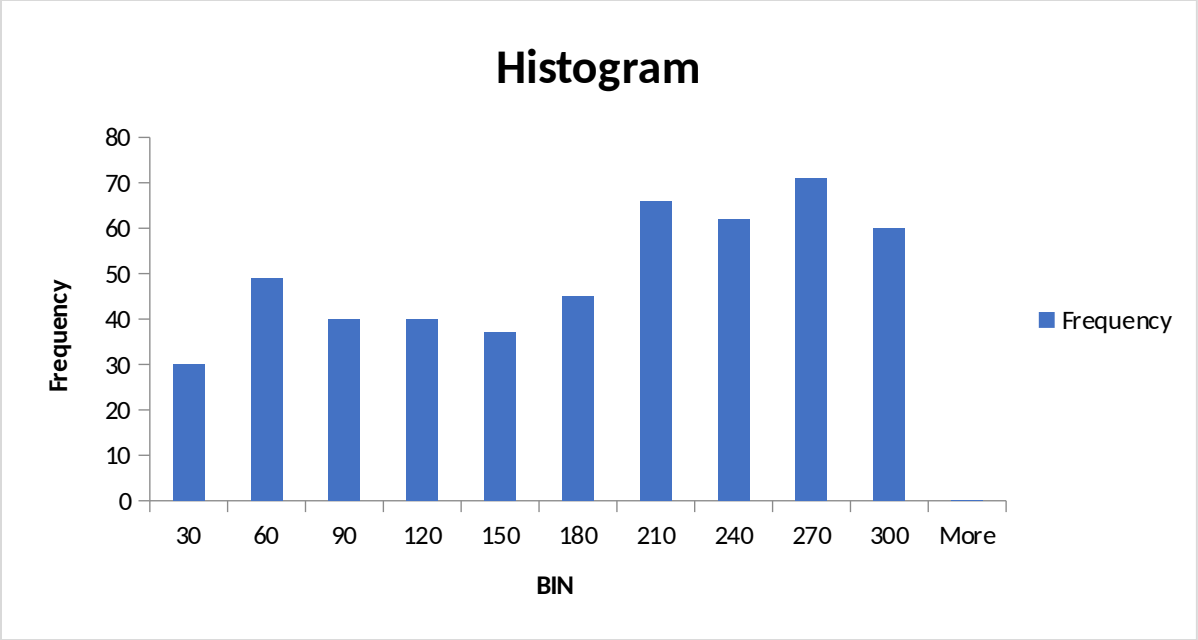


Step – 3 : Select raw data as 'Input Range' and select bin as 'Bin Range'. Select labels, new worksheet and chart output and click OK.



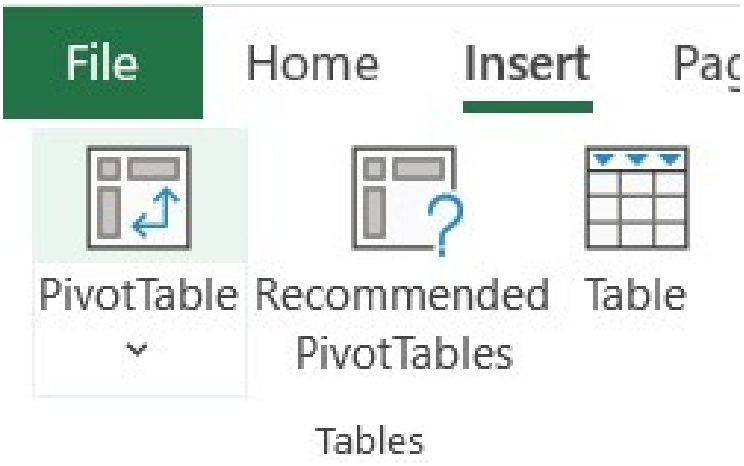
Step – 4 : Generated Result.

Bin (sec)	Frequency
30	30
60	48
90	40
120	40
150	37
180	45
210	66
240	62
270	71
300	60

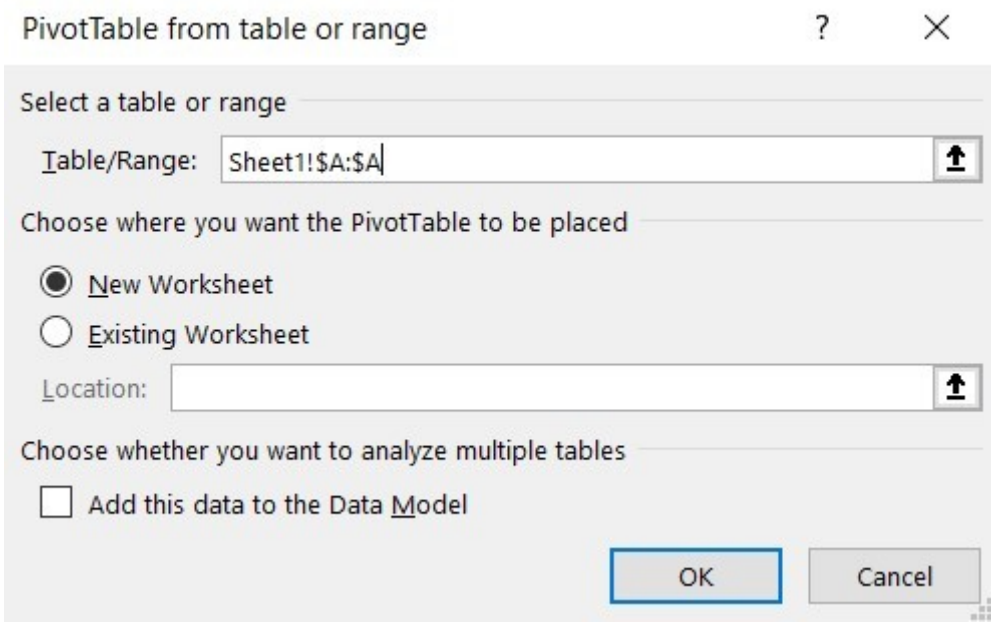


2. ***<u>Another way we use Pivot Table to calculate the Frequency table.</u>***

<u>Step – 1 :</u> Click on 'Insert' Tab and select 'Pivot Table' option.



Step – 2 : Select the raw data as Table/Range, select new worksheet and click OK.



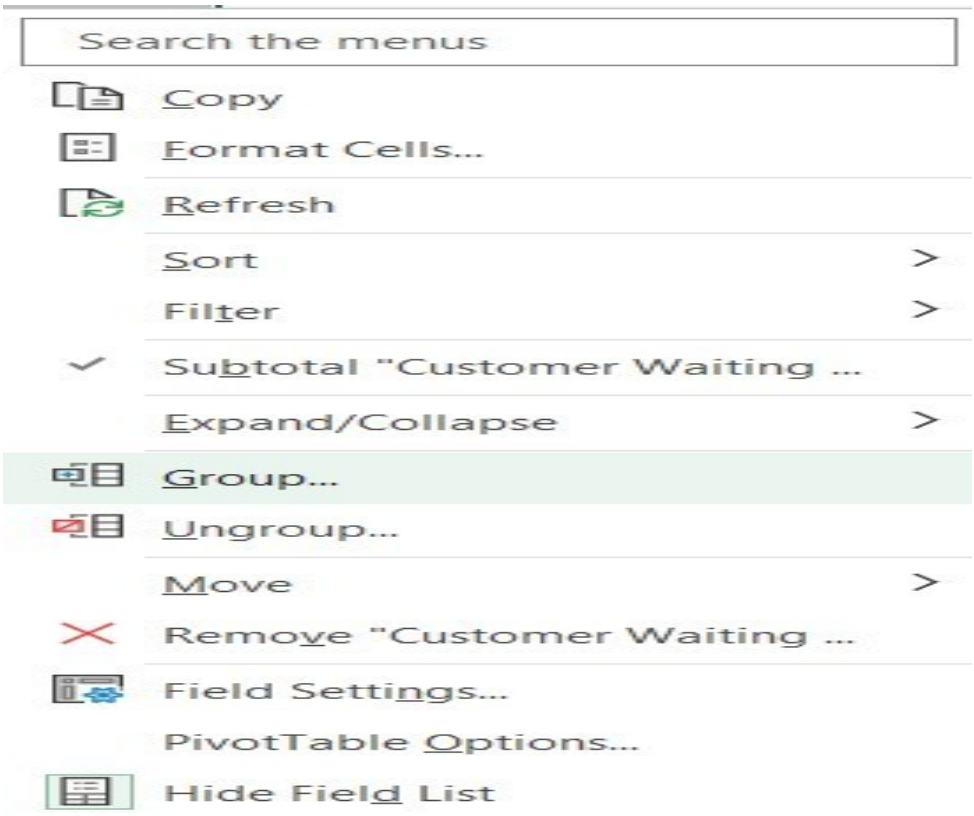
Step – 3 : Click and drag table to 'Rows' and 'Values' Section.



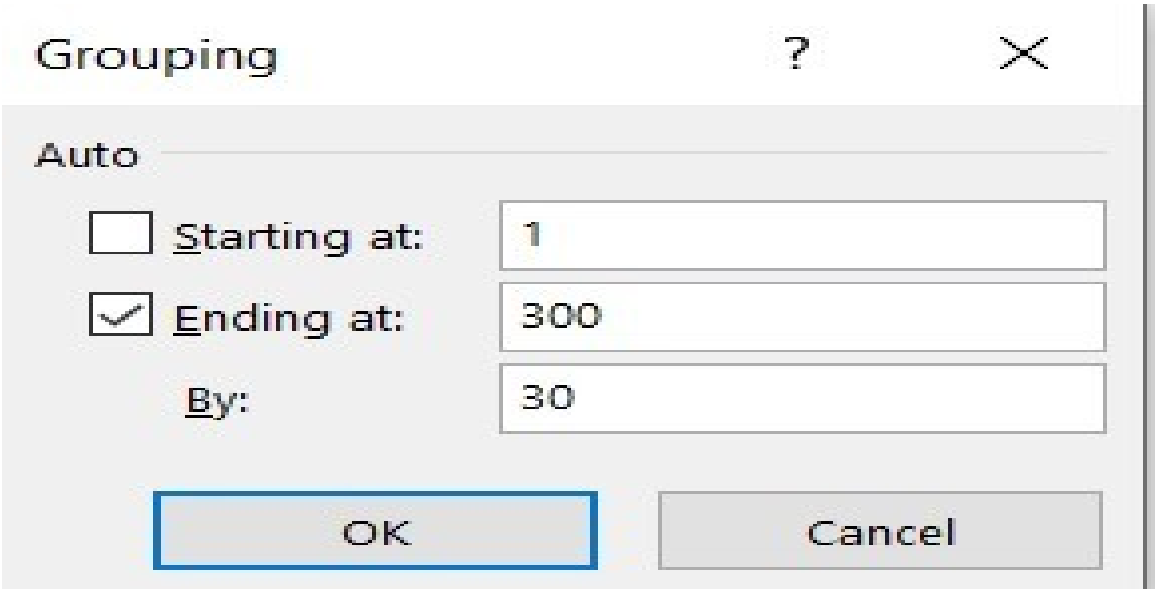
Step – 4 : To create Bin Range(class interval) we have to select the data under the column.

3	Row Labels	Count of Customer Waiting
4	0	
5	1	
6	2	
7	4	
8	6	
9	8	
10	9	
11	12	
12	15	
13	18	
14	19	

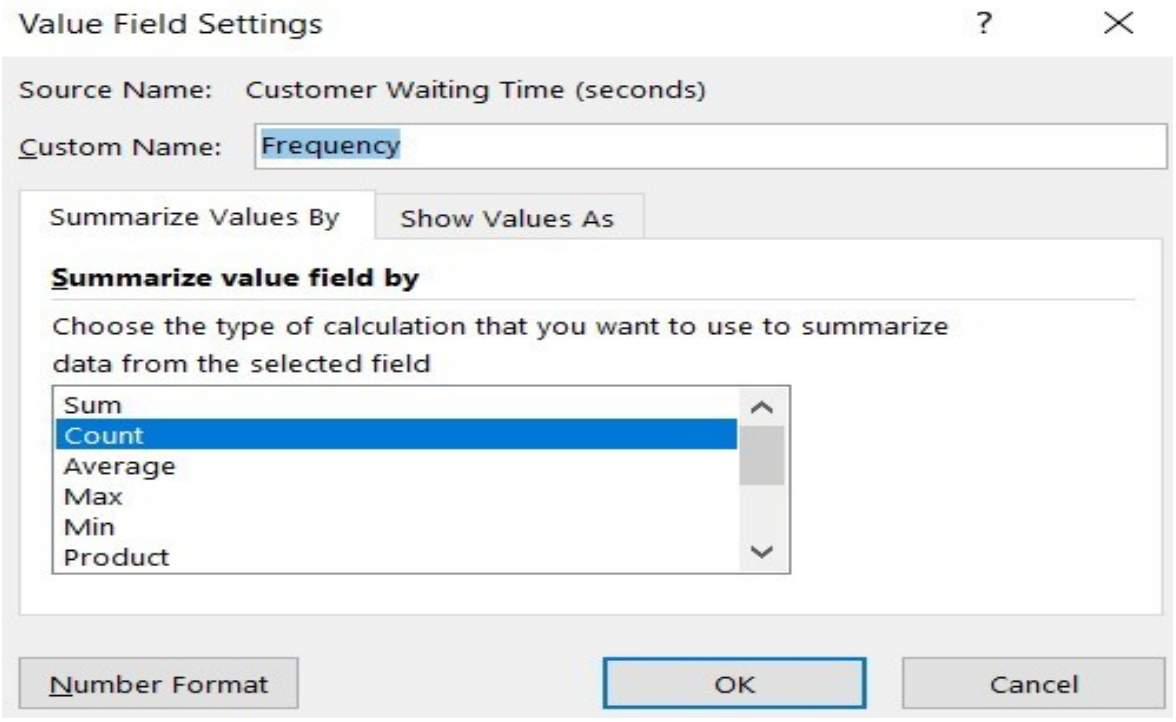
Step – 5 : Right click on the selected data and select 'Group' option from the menu.



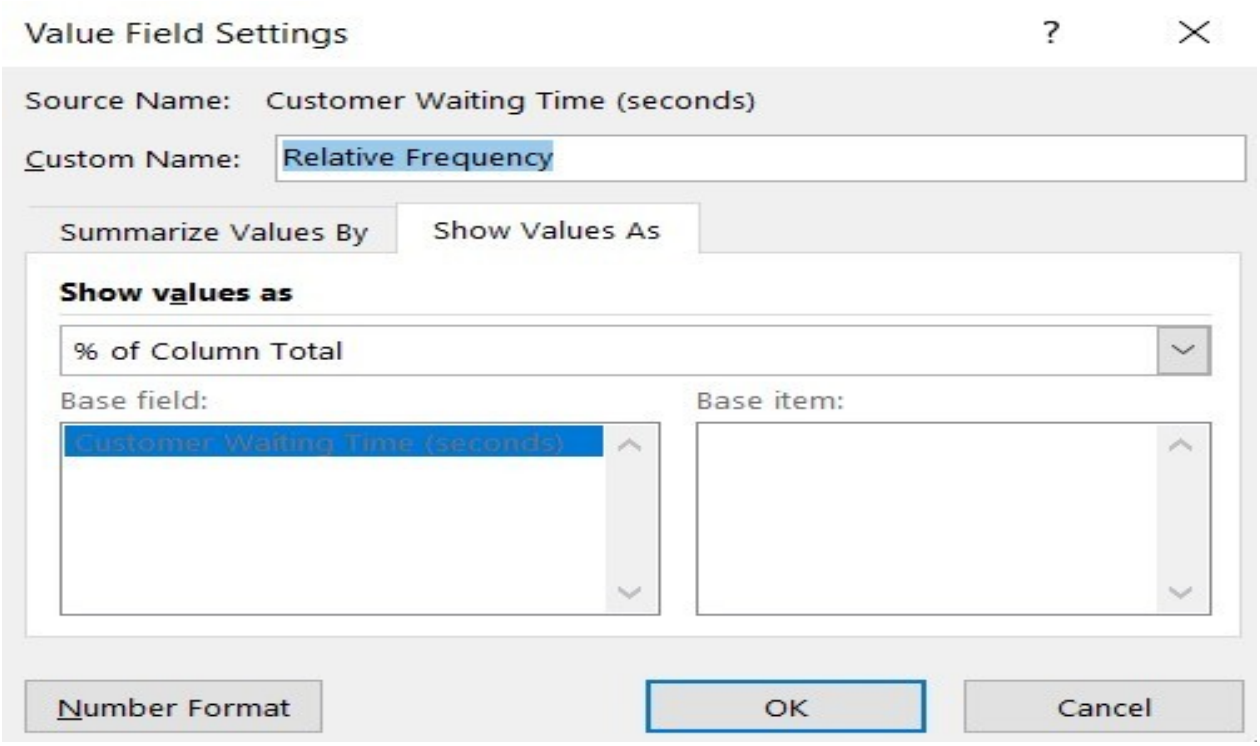
Step – 6 : Enter the following values and click OK.



Step – 7 : Rename the 'Frequency' column.



<u>Step – 8</u> : Copy the table into the another location and right click on the 'Frequency' column and click OK.



<u>Step – 9 :</u> Final Result.

Bin Range	Frequency
<1 or (blank)	2
1-30	28
31-60	49
61-90	40
91-120	40
121-150	37
151-180	45
181-210	66
211-240	62
241-270	71
271-300	60
Grand Total	**500**

Bin Range	Relative Frequency
<1 or (blank)	0%
1-30	6%
31-60	10%
61-90	8%
91-120	8%
121-150	7%
151-180	9%
181-210	13%
211-240	12%
241-270	14%
271-300	12%
Grand Total	**100%**

4. SUMMARY.

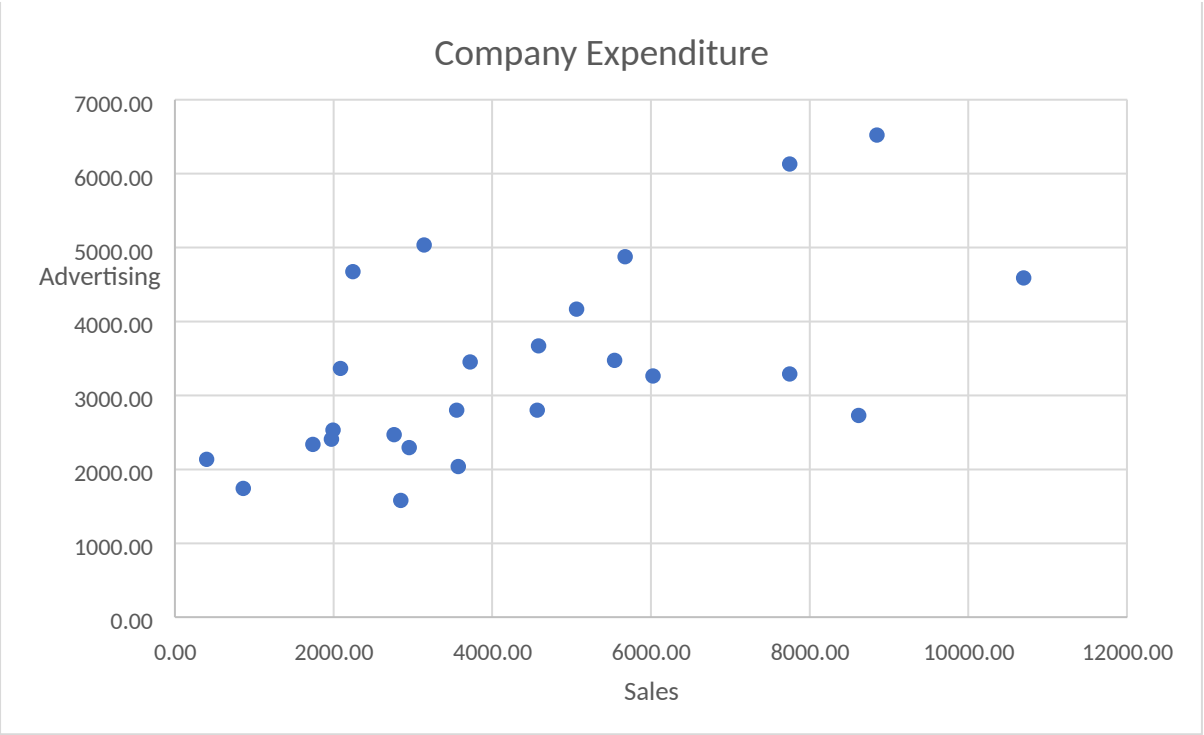
To find minimum we use **=min()** , to find maximum we have used **=max()** and for count**=count()**. To find frequency we need to use **=frequency(given data, bin range)** and to find total of frequencies we use **=sum(frequencies).** And to find relative frequency we use **=frequency/total of frequency.** It can also be plotted in shortcut by using data analysis toolpack and pivot table as shown.

ASSIGNMENT - 3

GIVEN DATA

Advertising	Sales
4582.88	3669.88
5539.78	3473.95
2950.38	2295.10
2243.07	4675.56
7747.08	6125.96
402.44	2134.94
3140.62	5031.66
2086.16	3367.45
8846.25	6519.45
5673.11	4876.37

Plot Scatter plot based on advertising and Sales.



1. Summary.

By Plotting scatter plot we can see that there is an increment of sales in the company in that year through advertisement.

ASSIGNMENT – 4

GIVEN DATA

Day	Date	New York
1	9-15-2015	74.5
2	9-14-2015	67.0
3	9-13-2015	70.5
4	9-12-2015	74.5
5	9-11-2015	73.5
6	9-10-2015	74.5
7	9-9-2015	82.5
8	9-8-2015	85.5
9	9-7-2015	80.5
10	9-6-2015	75.5
11	9-5-2015	73.5
12	9-4-2015	77.0
13	9-3-2015	84.5
14	9-2-2015	81.5
15	9-1-2015	82.0

1. Finding central tendencies after using functions.

We've to find the mean, median, mode and standard deviation of the given data. And to get these central tendencies we've to follow the steps as follows.

To find **mean**, we should use **=average()**

To find **median**, we should use **=median()**

To find **mode**, we should use **=mode()**

To find **variance**, the user should use **=var.s()**

To find **standard deviation**, the user should use **=stdev.s()**

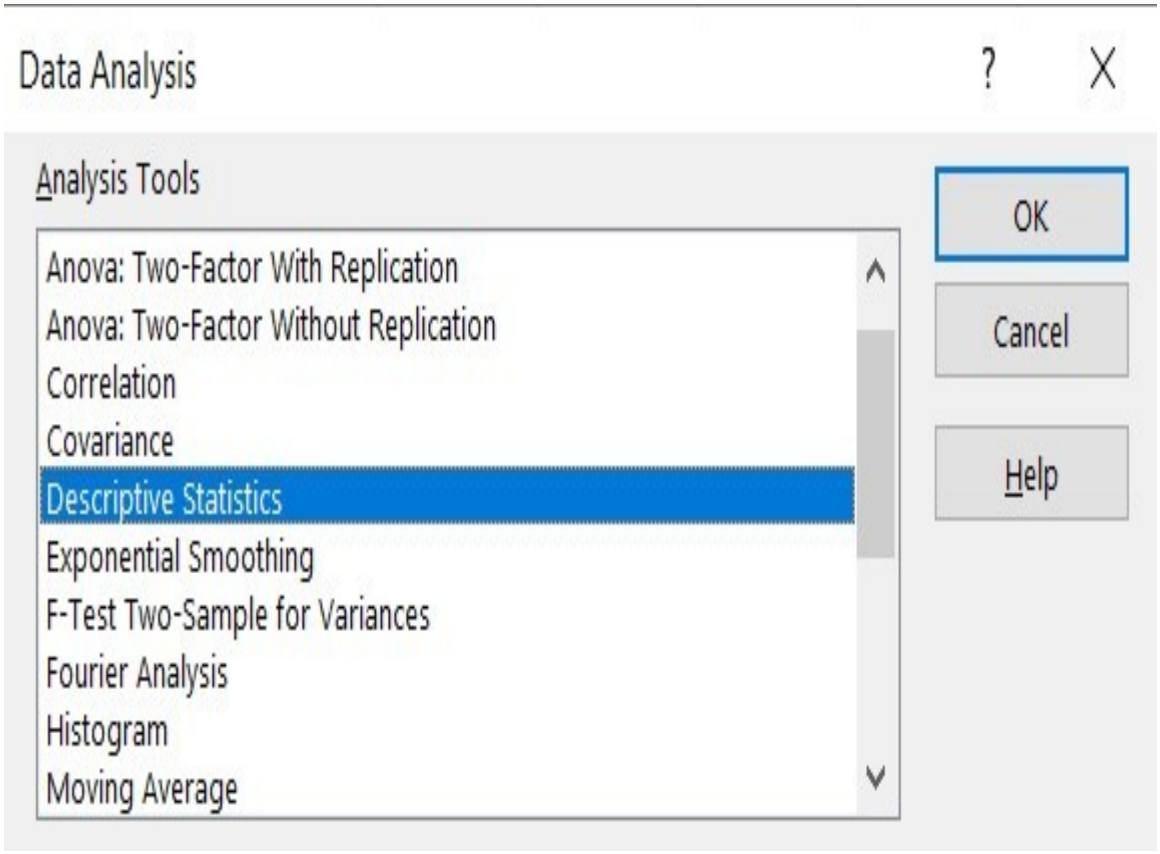
2. Finding central tendencies using data analysis toolpak.

Step – 1 : Select 'Data Analysis' from 'Analysis' section of 'Data' tab.

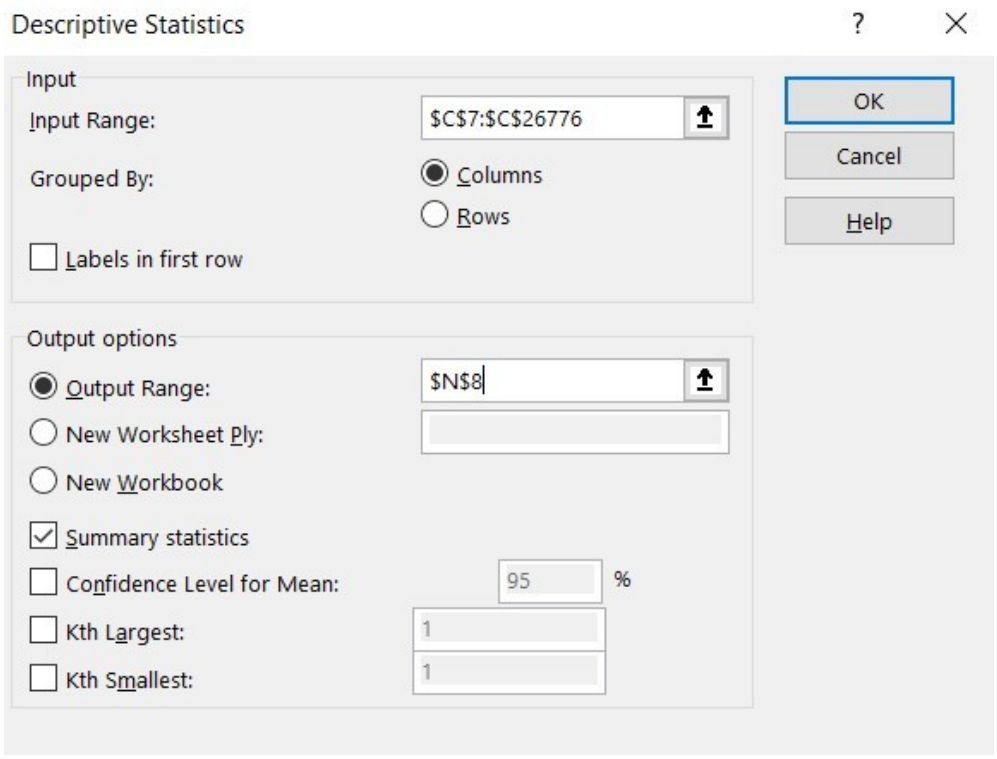




Step – 2 : Select 'Descriptive Statistics' and click OK.



<u>Step – 3 :</u> Select raw data as 'Input Range', select the radio button of the 'Output Range', select 'Summary statistics' and click OK.



<u>Step – 4</u> : Generated Result.

Mean	55.23240232
Standard Error	0.106222471
Median	55.94
Mode	76.01
Standard Deviation	17.37963233
Sample Variance	302.05162
Kurtosis	-0.932232962
Skewness	-0.185382888
Range	91.89
Minimum	2.57
Maximum	94.46
Sum	1478571.41
Count	26770

Summary.

The data given to us is the data of the temperature (in Fahrenheit) of New York City from 1st June 1942 to 15th September 2015 where we can see that the average temperature of the city has been rising from decades to decades.

*The analysers can use the certain methods as mentioned before to get the descriptive central tendencies for their raw data by applying functions and in shortcut of using the data analysis toolpak by which we found the **mean** is 55.2, **median** is 55.9, **mode** is 76.01, **variance** is 302.0516, **standard deviation** is 17.37963.*

ASSIGNMENT - 5

1. **A) Find the variance and SD for the given data.**

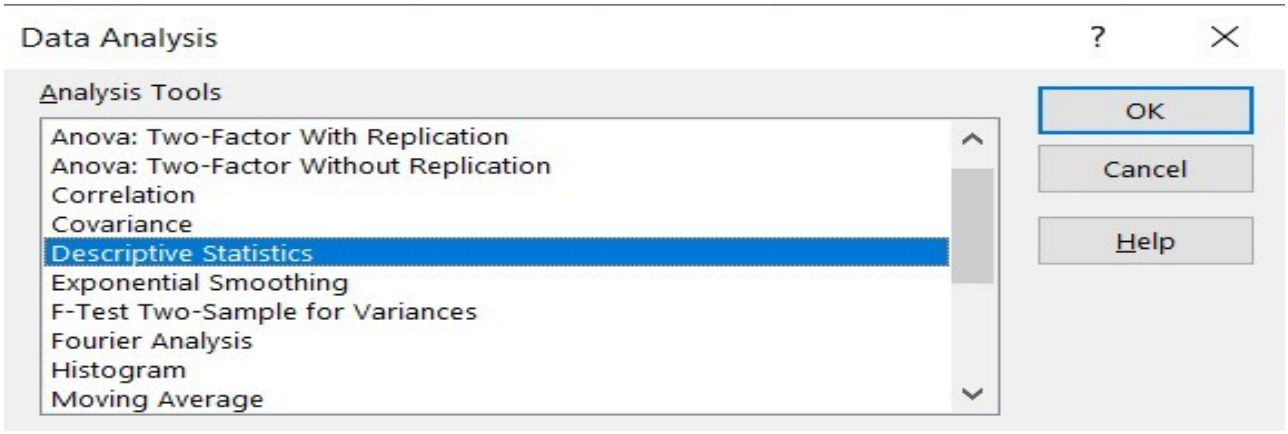
Student	Score	Difference	Difference^2	Variance	Standard Deviation
Daniela	85	0.7	0.49	40.45555556	6.360468187
Tommy	79	-5.3	28.09		
Edward	90	5.7	32.49		
Julia	88	3.7	13.69		
Timothy	88	3.7	13.69		
Peter	75	-9.3	86.49		
Neal	92	7.7	59.29		
Sally	74	-10.3	106.09		
Mike	83	-1.3	1.69		
Adam	89	4.7	22.09		
Total	**843**		**364.1**		
Count	**10**				
N-1	**9**				
Average	**84.3**				

B) Summary statistics using data analysis toolpack.

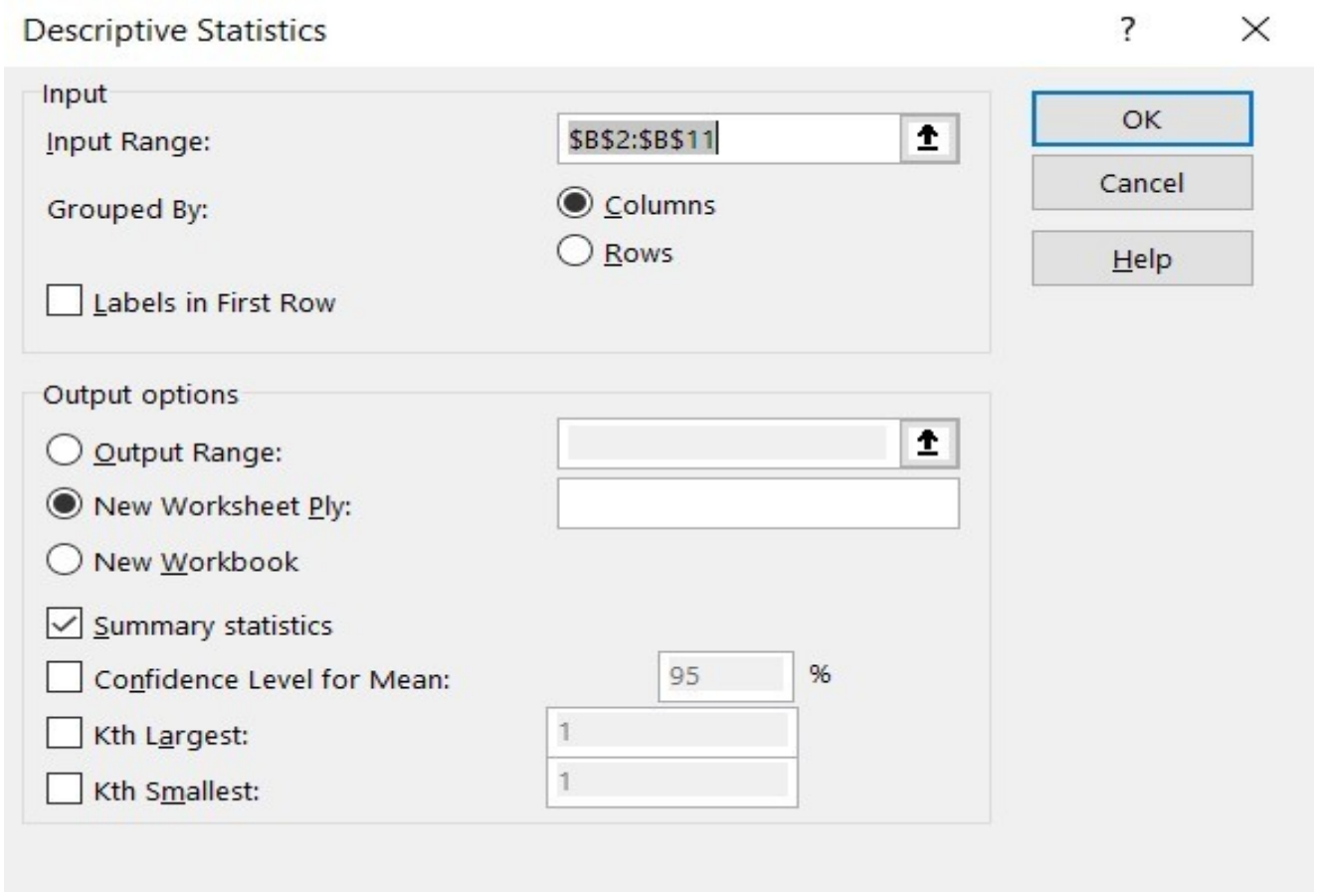
Step – 1 : Select “Data Analysis” from Data tab.



<u>Step 2</u>: Select “Desccriptive Statistics From Data Analysis Tool Pack



Step – 3 : Select 'Score' as 'Input Range', select the radio button of the 'New Worksheet', select 'Summary statistics' and click OK.



Student	Score	Difference	Difference^2	Variance	Standard Deviation
Daniela	85	1	1	57.80952381	7.603257447
Tommy	79	-5	25		
Edward	90	6	36		
Julia	88	4	16		
Timothy					
Peter	75	-9	81		
Neal	92	8	64		
Sally	74	-10	100		
Mike					
Adam	89	5	25		
Total	**672**		**348**		
Count	**8**				
N-1	**7**				

Average	**84**				

Step 4:

Final Result.

Mean	84.3
Standard Error	2.011356646
Median	86.5
Mode	88
Standard Deviation	6.360468187
Sample Variance	40.45555556
Kurtosis	-1.010577396
Skewness	-0.648229501
Range	18
Minimum	74
Maximum	92
Sum	843
Count	10

2. A) Find the variance and SD for the given data with missing values.

B) Summary statistics using data analysis toolpak.

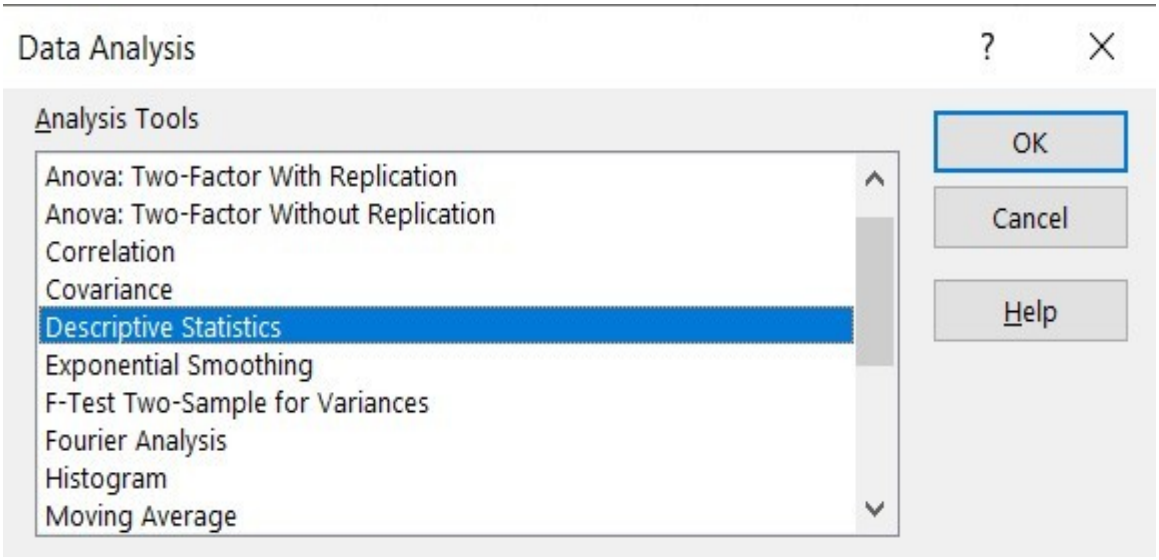
B) Summary statistics using data analysis toolpack

Step – 1 : Select 'Data Analysis' from 'Analysis' section of 'Data' tab.

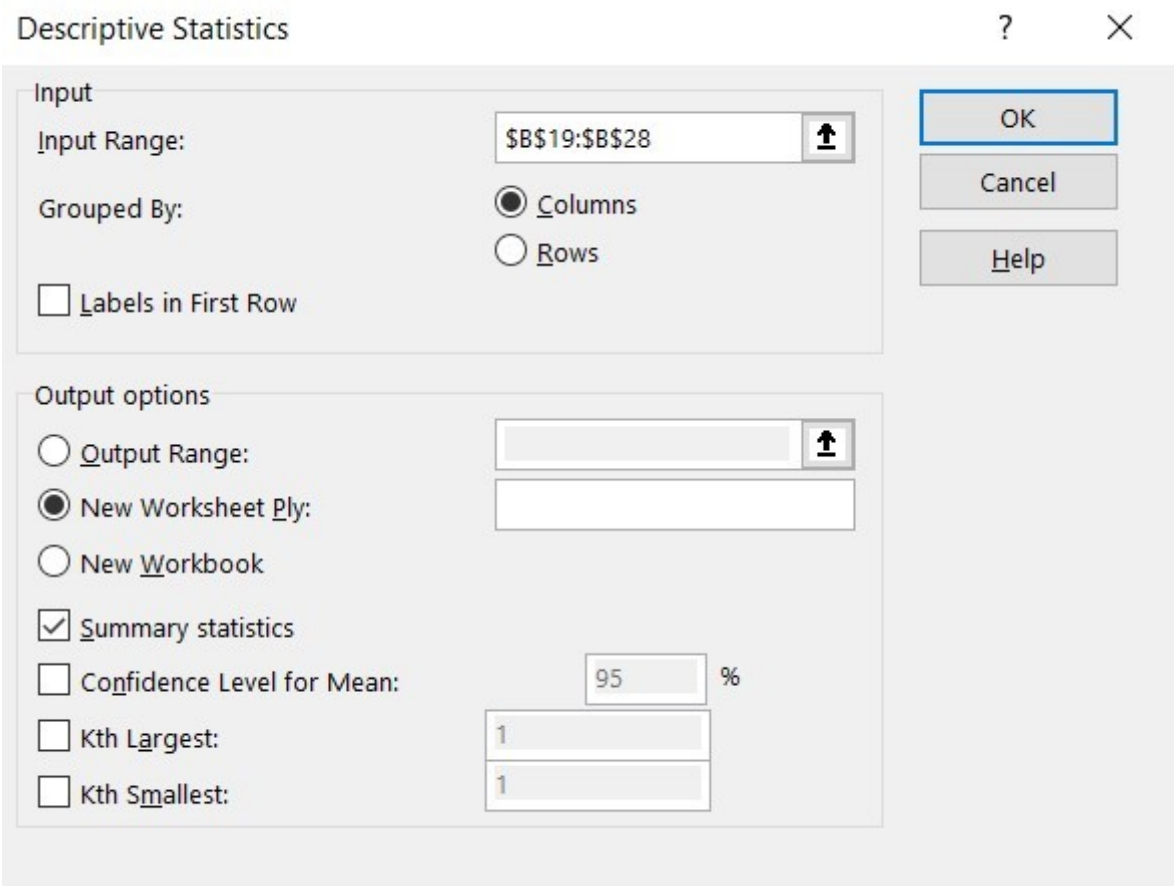




Step – 2 : Select 'Descriptive Statistics' and click OK.



Step 3: Select raw data as 'Input Range', select the radio button of the 'New Worksheet' and click OK.



Step – 4 : Generated Result.

Mean	84
Standard Error	2.49284691
Median	86.5
Mode	#N/A
Standard Deviation	7.050835817
Sample Variance	49.71428571
Kurtosis	-1.645541023
Skewness	-0.508621529
Range	18
Minimum	74
Maximum	92
Sum	672
Count	8

3. **Summary.**

SUMMARY

Here for count we can use **=countif()** for count**, n-1** ,for mean average we can use **=average() =difference^2** and finding variance and standard deviation by **=sum_of_difference/N-1** and **=sqrt(variance).** All this can be found in shortcut way of using data analysis toolpack as mentioned steps before.

In the first complete and proper data set we can see the variance of 40.45555556 and standard deviation of 6.360468187 and so goes with the summary of data analysis toolpak. Whereas we can see in the second data set with few missing values which also makes a proper variance of 57.80952381 and standard deviation of 7.603257447 and so goes with the summary of data analysis toolpak.

ASSIGNMENT – 6

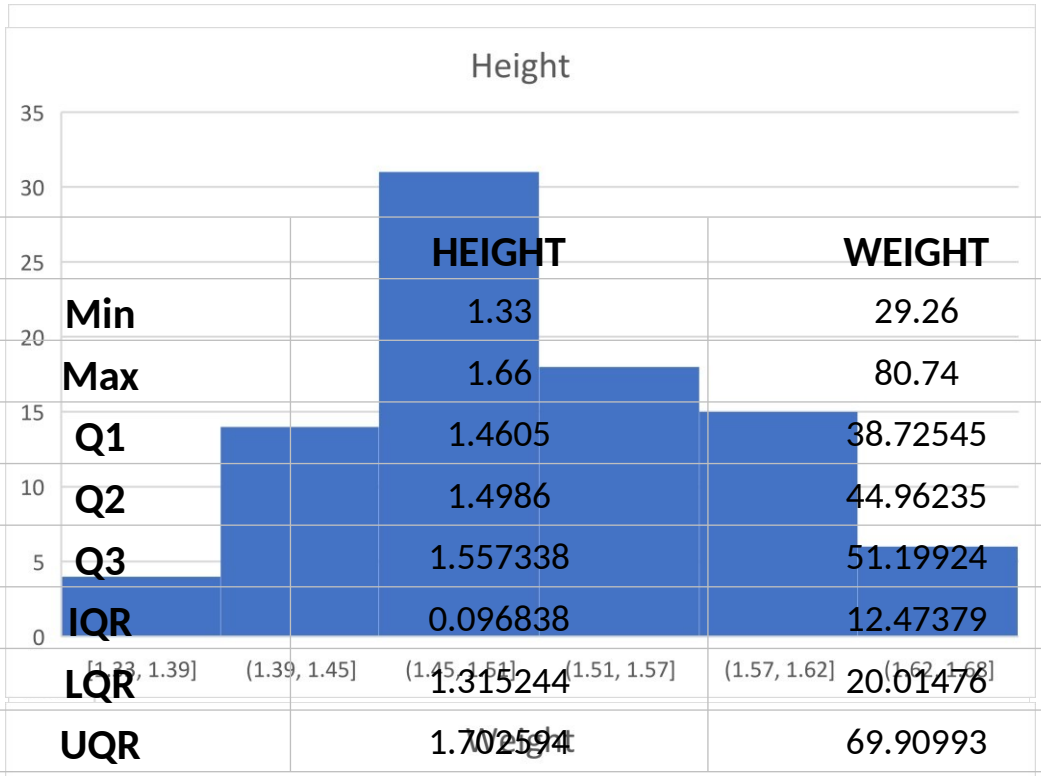
Height M	Weight kg
1.60	49.44
1.65	62.60
1.65	75.75
1.53	48.99
1.45	43.09
1.61	52.62
1.56	47.97
1.50	45.59
1.52	47.85
1.48	44.45
1.47	46.04
1.55	53.07

1.52	65.88
1.54	46.04
1.46	43.54

RAW DATA (sample data)

1. **Find variability using QR of the given data of a class students height and weight.**

2. **Plot Box & Whisker graph.**
3. **Comparison of histograms.**



	HEIGHT	WEIGHT
Min	1.33	29.26
Max	1.66	80.74
Q1	1.4605	38.72545
Q2	1.4986	44.96235
Q3	1.557338	51.19924
IQR	0.096838	12.47379
LQR	1.315244	20.01476
UQR	1.702594	69.90993

4.SUMMARY STATISTICS USING DATA ANALYSIS TOOLPACK

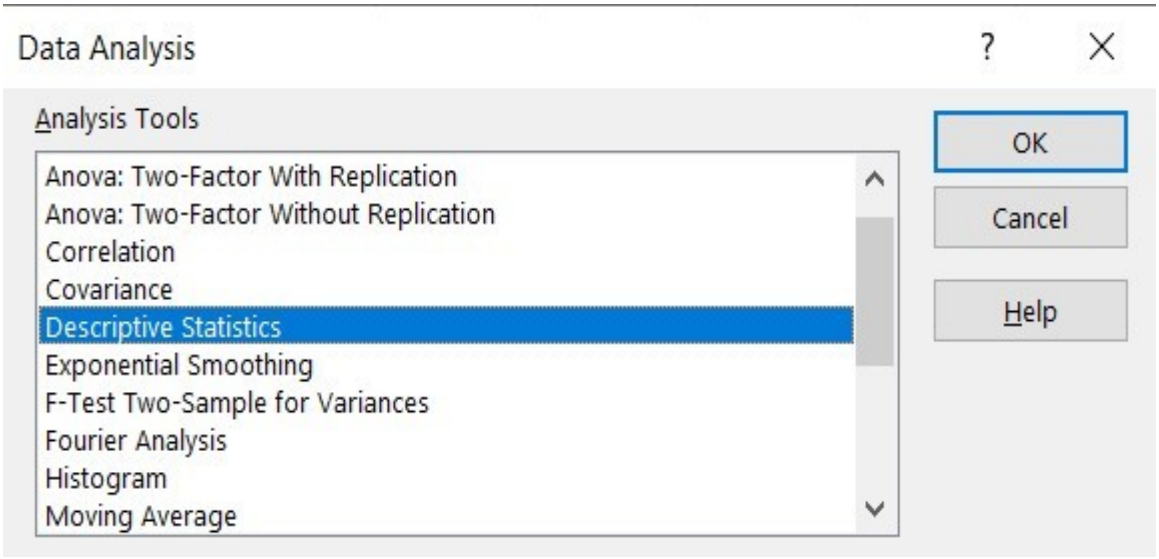
For the data of Height.

Step – 1 : **Select 'Data Analysis' from 'Analysis' section of 'Data' tab.**



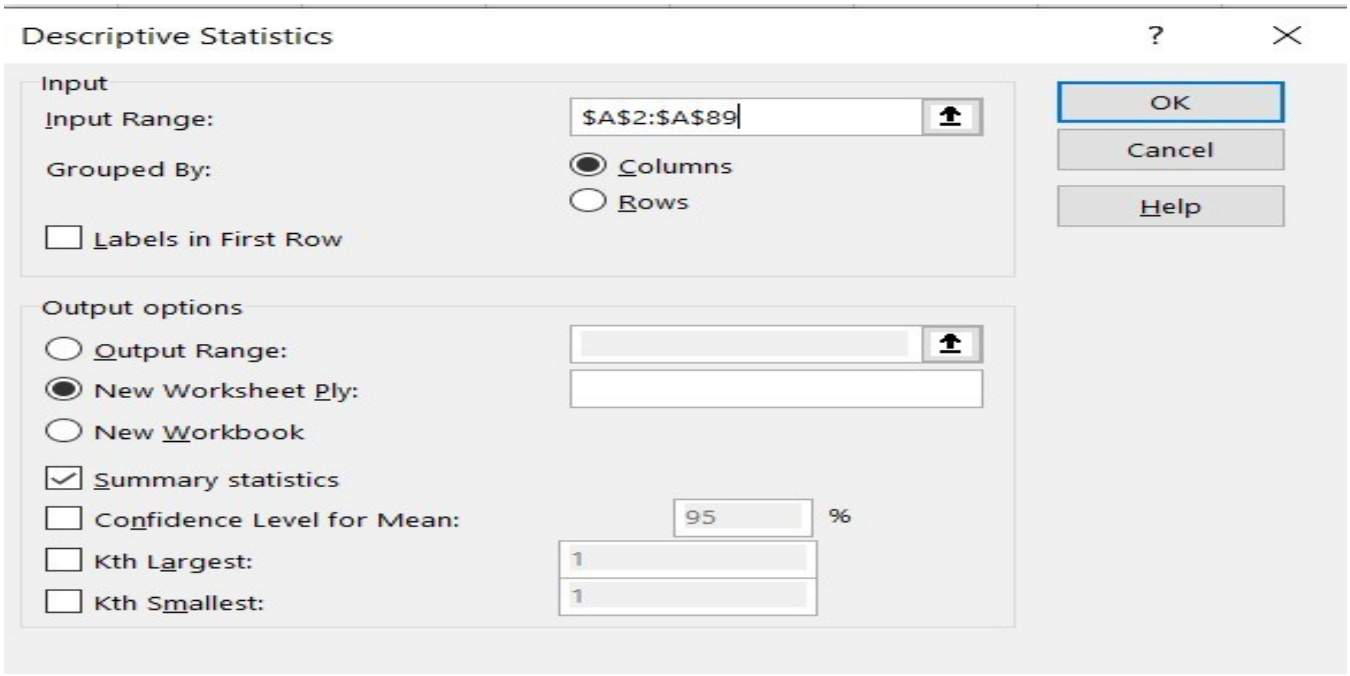


Step – 2 : Select 'Descriptive Statistics' and click OK.



Height (M)	
Mean	1.508413636
Standard Error	0.007813553
Median	1.4986
Mode	1.4732
Standard Deviation	0.073297621
Sample Variance	0.005372541
Kurtosis	-0.34564739
Skewness	0.114950996
Range	0.3302
Minimum	1.3335
Maximum	1.6637
Sum	132.7404
Count	88
Largest(1)	1.6637
Smallest(1)	1.3335
Confidence Level(95.0%)	0.015530282

Step 3: Raw data as input range ,select the radio button of the select 'New Worksheet' and Summary statistics and click OK.



Step – 4 : Generated Result.

Height M	
Mean	1.508413636
Standard Error	0.007813553
Median	1.4986
Mode	1.4732
Standard Deviation	0.073297621
Sample Variance	0.005372541
Kurtosis	-0.34564739
Skewness	0.114950996
Range	0.3302
Minimum	1.3335
Maximum	1.6637
Sum	132.7404
Count	88
Largest(1)	1.6637
Smallest(1)	1.3335
Confidence Level(95.0%)	0.015530282

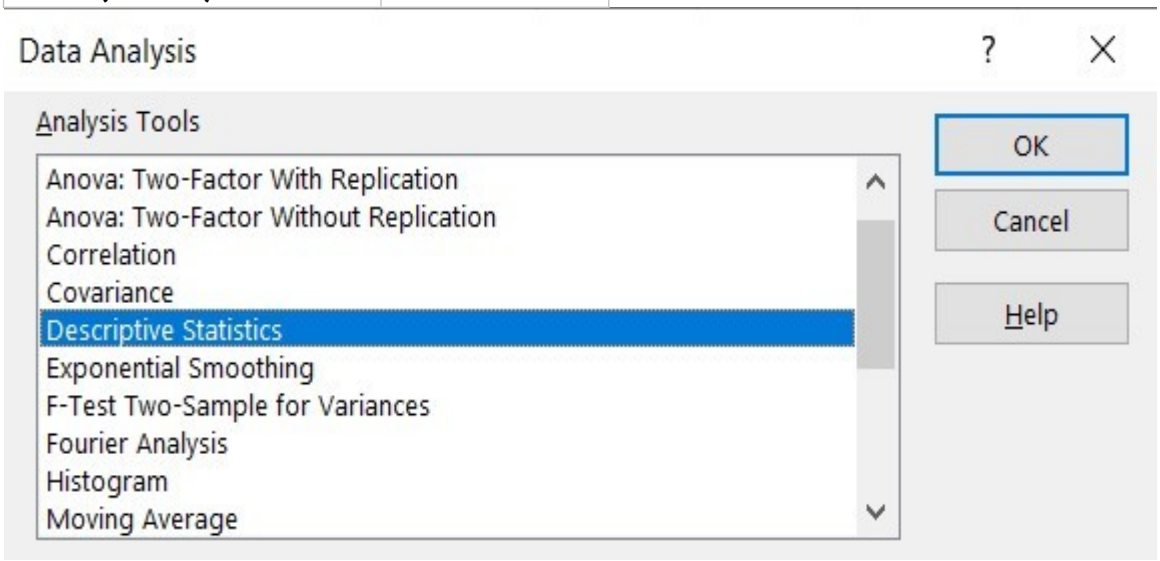
Step – 1 : Select ‘Data Analysis’ from ‘Analysis’ section of ‘Data’ tab.

Weight (KG)	
Mean	46.32698969
Standard Error	1.180230696
Median	44.96234665
Mode	47.8539982
Standard Deviation	11.07154531
Sample Variance	122.5791156
Kurtosis	1.145426086
Skewness	1.0524857
Range	51.4827374
Minimum	29.2567098
Maximum	80.7394472
Sum	4076.775093
Count	88
Largest(1)	80.7394472
Smallest(1)	29.2567098
Confidence Level(95.0%)	2.345836305

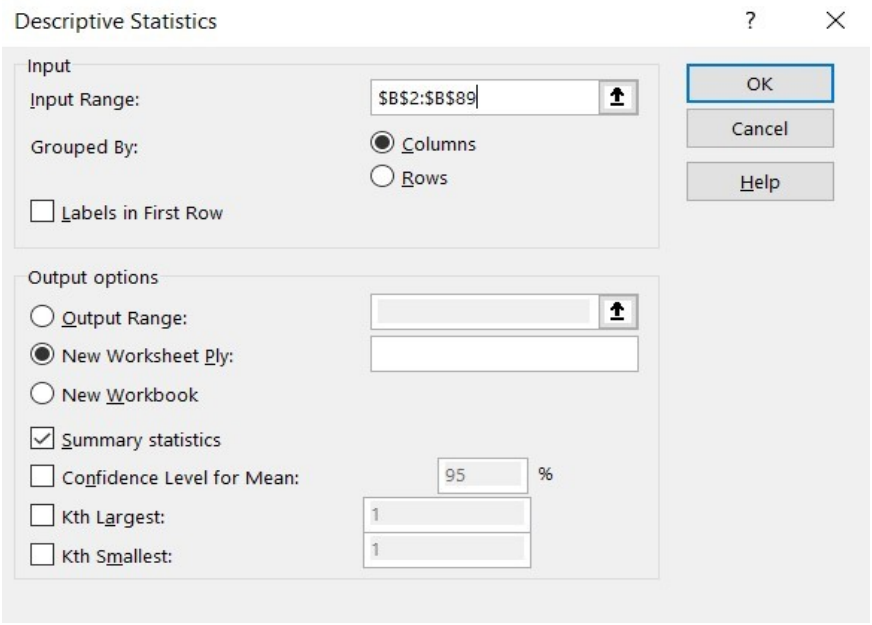




Step – 2 : Select 'Descriptive Statistics' and click OK.



Step – 3 : Select raw data as 'Input Range', select the radio button of the 'New Worksheet' and click OK.



Step – 4 : Generated Result.

5. Summary.

SUMMARY

Here we can find minimum unit and maximum unit by using inbuilt functions such as ***=min()*** *and* ***=max()****, Q1,Q2 and Q3 by using inbuilt functions such as* ***=quartile.inc(range,1); =quartile.inc(range,2); =quartile.inc(range,3)****, IQR by* ***=Q3-Q1****, and LQR and UQR by using* ***=Q1-(IQR*1.5)*** *and* ***=Q3+(IQR*1.5)****.*

The descriptive statistics of the data in shortcut by using data analysis toolpak as mentioned before.

In height, we can see that the minimum height of the class students is 1.33 with the maximum height of the class students is 1.66 where Q1 = 1.4605, Q2 = 1.4986 and Q3 = 1.557338 with the IQR of 0.096838 and LQR of 1.315244 and UQR of 1.702594.

ASSIGNMENT – 8

Exploratory data using Excel.

RegNO:- 231BGADA22
Name:- Anthony christopher P

Assignment

1) A Bank manager wants to get a senses of how many people are working to see a seller and conduct study & find the following pattern.

No of Customers waiting time	No of times observed
0	3
1	10
2	8
3	5
4	3
5	2
6	1

(i) What is the Expected no of customers waiting

No of customers	No of time observed	Probablity P(x=x)	Cummulative Probablity
0	03	3/32 = 0.0975	0.09375
1	10	0.3125	0.40625
2	08	0.25	0.65625
3	05	0.15625	0.8125
4	03	0.09675	0.90625
5	02	0.0625	0.96875
6	01	0.03125	01

→ Expected no of customers waiting in the line

$E(x) = (0 \times 0.9375) + (1 \times 0.3125) + (2 \times 0.85) + (3 \times 0.15625) + (4 \times 0.09375) + (5 \times 0.0625) + (6 \times 0.03125)$

$= 2.15625$

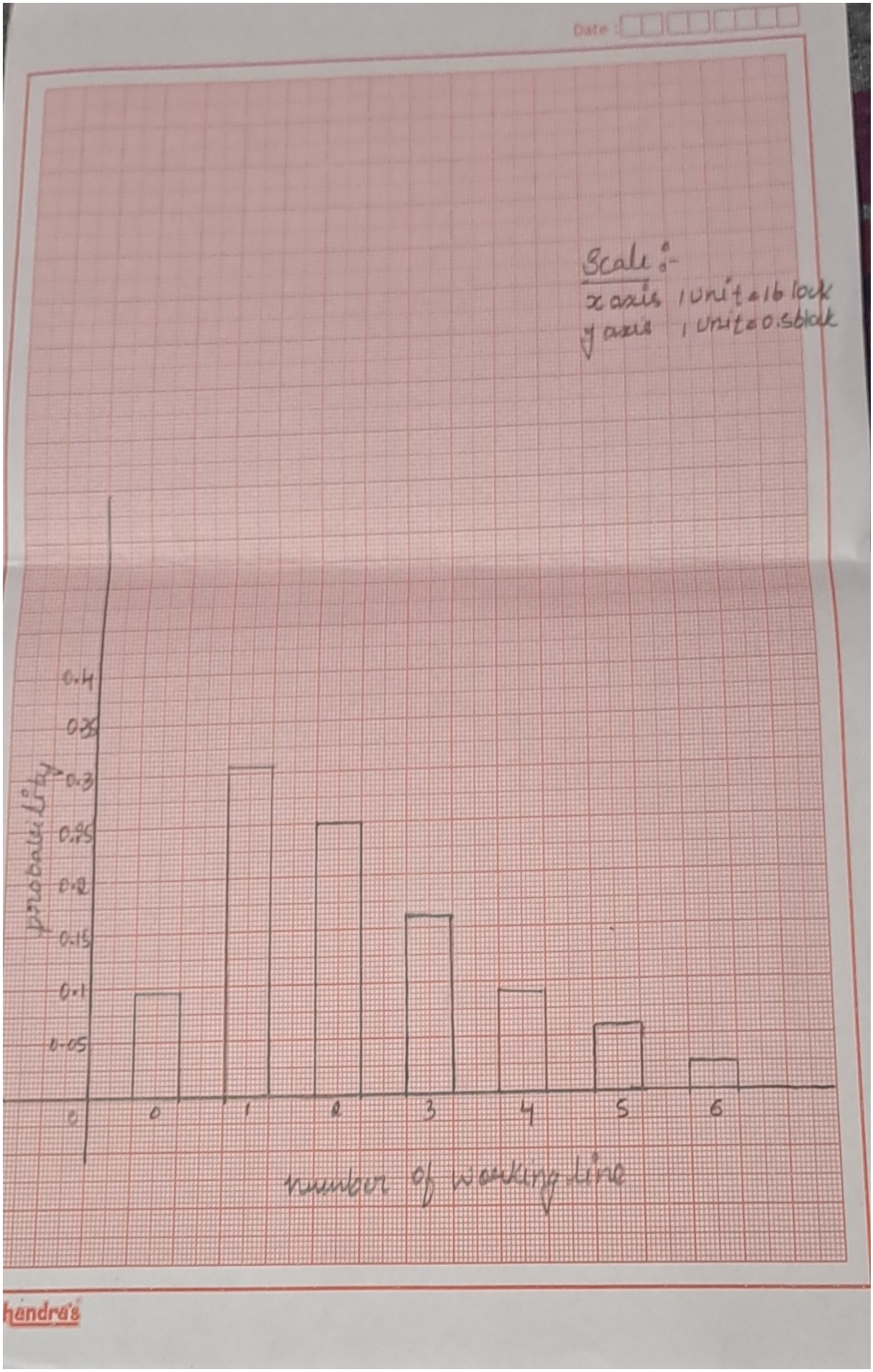
Expected no of customers waiting = 2.15625

ii) What is the probability to find 4 or more.

→ Probability $P(x \geq 4) = P(x=4) + P(x=5) + P(x=6)$

$= (0.09375) + (0.0625) + (0.03125)$

$= 0.1875$


Date :
Scale :-
x axis 1 unit = 1 block
y axis 1 unit = 0.5 block
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
probability
0
1
2
3
4
5
6
number of working days

ASSIGNMENT – 9

Daily Demand	Occurrences	Probability	(x-mu)^2
1	3	0.021428571	109.9502041
2	10	0.071428571	89.97877551
3	10	0.071428571	72.00734694
4	5	0.035714286	56.03591837
5	2	0.014285714	42.0644898
6	4	0.028571429	30.09306122
7	14	0.1	20.12163265
8	2	0.014285714	12.15020408
9	9	0.064285714	6.17877551
10	5	0.035714286	2.207346939
11	4	0.028571429	0.235918367
12	8	0.057142857	0.264489796
13	2	0.014285714	2.293061224
14	10	0.071428571	6.321632653
15	5	0.035714286	12.35020408
16	3	0.021428571	20.37877551
17	4	0.028571429	30.40734694
18	13	0.092857143	42.43591837
19	15	0.107142857	56.4644898
20	12	0.085714286	72.49306122
	140	**1**	

1. **The manager of a convenience store wants to understand the demand for one of the products she sells, so that she can make a better decision on how many to stock every day. She has the following data gathered over several weeks. Based on this data, what is the value of expected demand and its standard deviation?**

Sum	140
Expected Mean	11.48571429
Standard Deviation	6.20539594
Expected Mean + Standard Deviation	17.69111023
Expected Mean - Standard Deviation	5.280318345

SUMMARY

We can find expected mean by using function **=sumproduct(daily demand, probability)** and standard deviation by function **=sqrt(sumproduct(probability, (x-mu)^2)).**

ASSIGNMENT – 10

Height M	Weight kg
1.60	49.44
1.65	62.60
1.65	75.75
1.53	48.99
1.45	43.09
1.61	52.62
1.56	47.97
1.50	45.59
1.52	47.85
1.48	44.45
1.47	46.04
1.55	53.07
1.52	65.88
1.54	46.04
1.46	43.54
1.52	62.37
1.46	45.81
1.58	74.39
1.52	55.57
1.50	46.15
1.49	47.85
1.48	42.18
1.59	45.81
1.56	44.68
1.58	42.64
1.57	43.54
1.50	37.31
1.52	39.35
1.44	39.01
1.49	40.60

Here we're going to apply Karl Pearson's correlation method on few data sets such as following:-

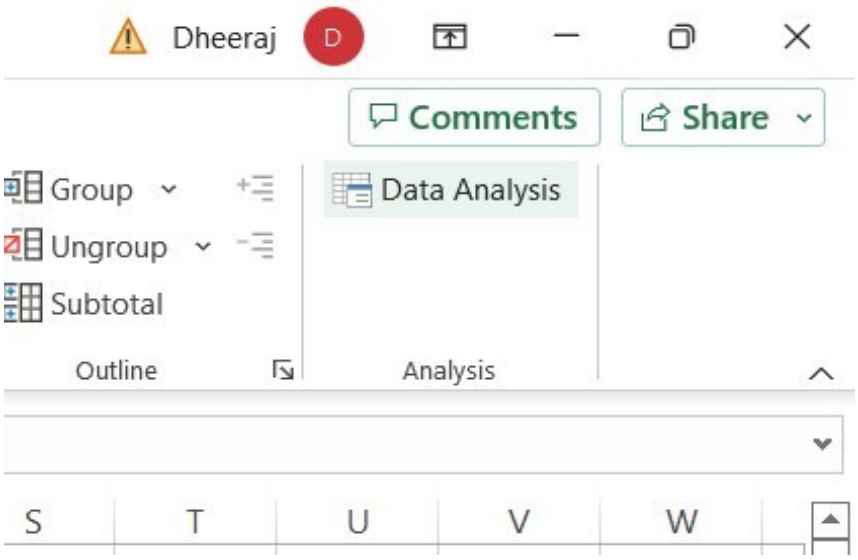
The data below is a 2 variable distribution and numeric in nature.

Actually due to vast distribution we took sample data of height and weight of 30 students out of the population data of 88 students.

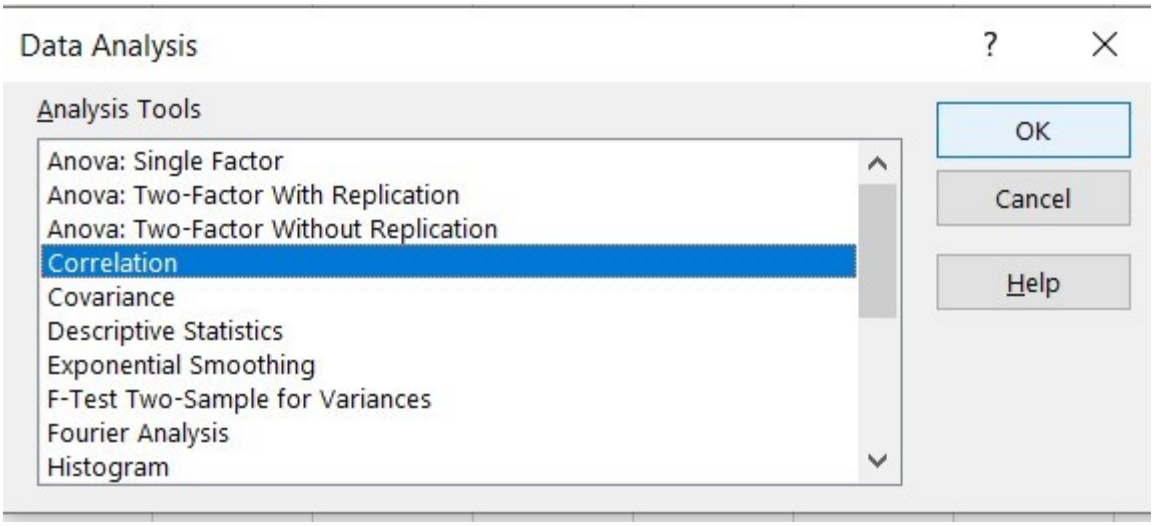
To perform correlation on the whole distribution, there are certain steps to be followed using data analysis tool pack using following steps

.

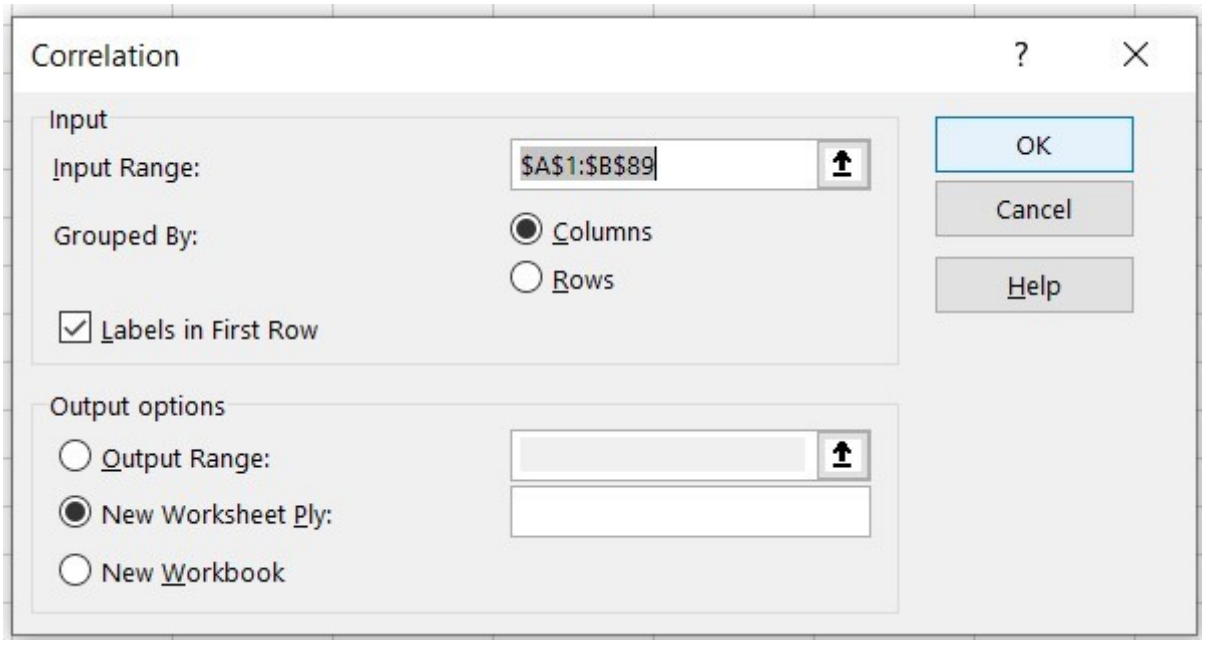
<u>Step – 1 : Go to 'Data tab' and select 'Data analysis' from 'Analysis' section.</u>



Step – 2 : Select 'correlation' and click OK.



Step – 3 : Select raw data in 'input range' section, select 'labels in the first row' and select 'New Worksheet Ply' and click OK.

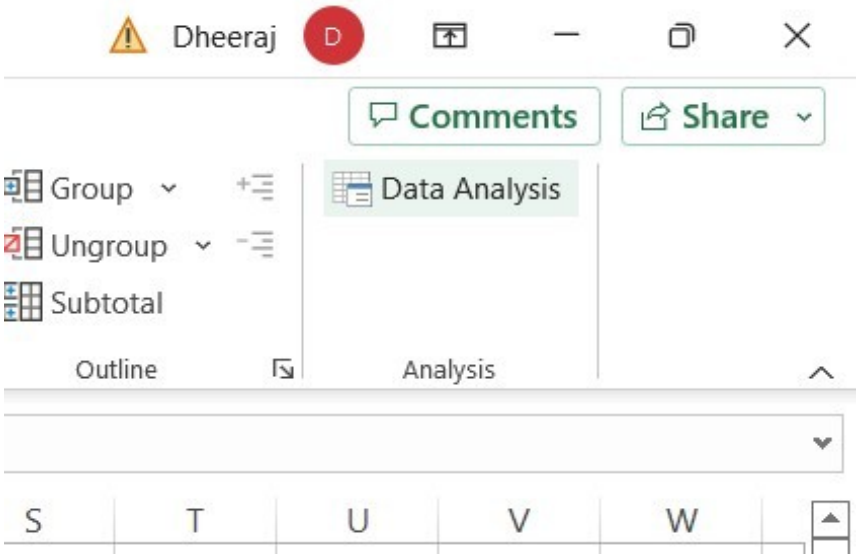


Step – 4 : Generated result.

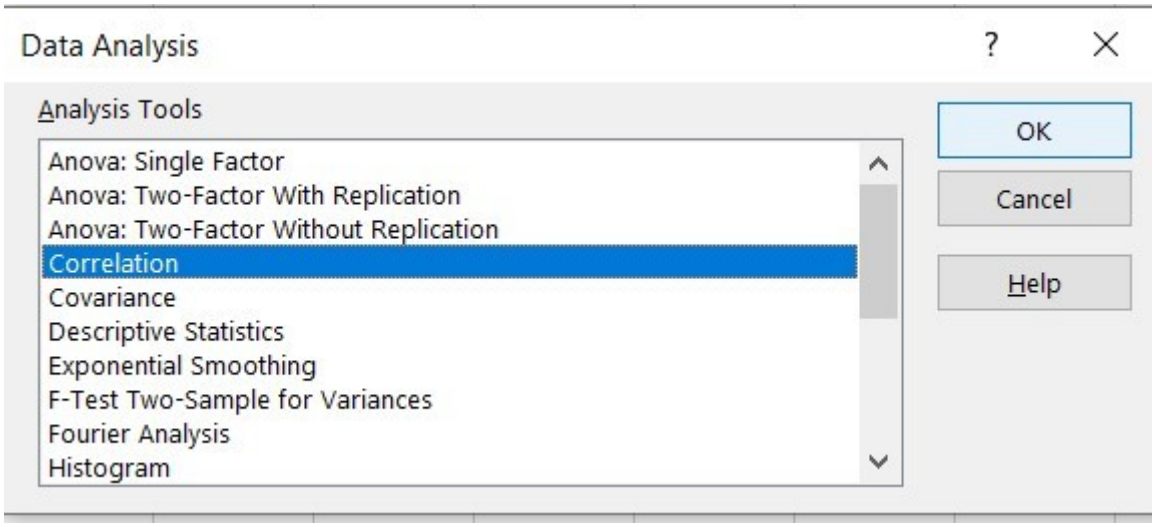
	Height M	*Weight kg*
Height M	1	
Weight kg	0.705099	1

age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
52	1	0	125	212	0	1	168	0	1	2	2	3	0
53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
61	1	0	148	203	0	1	161	0	0	2	1	3	0
62	0	0	138	294	1	1	106	0	1.9	1	3	2	0
58	0	0	100	248	0	0	122	0	1	1	0	2	1
58	1	0	114	318	0	2	140	0	4.4	0	3	1	0
55	1	0	160	289	0	0	145	1	0.8	1	1	3	0
46	1	0	120	249	0	0	144	0	0.8	2	0	3	0
54	1	0	122	286	0	0	116	1	3.2	1	2	2	0
71	0	0	112	149	0	1	125	0	1.6	1	0	2	1
43	0	0	132	341	1	0	136	1	3	1	0	3	0
34	0	1	118	210	0	1	192	0	0.7	2	0	2	1
51	1	0	140	298	0	1	122	1	4.2	1	3	3	0
52	1	0	128	204	1	1	156	1	1	1	0	0	0
34	0	1	118	210	0	1	192	0	0.7	2	0	2	1
51	0	2	140	308	0	0	142	0	1.5	2	1	2	1
54	1	0	124	266	0	0	109	1	2.2	1	1	3	0
50	0	1	120	244	0	1	162	0	1.1	2	0	2	1
58	1	2	140	211	1	0	165	0	0	2	0	2	1

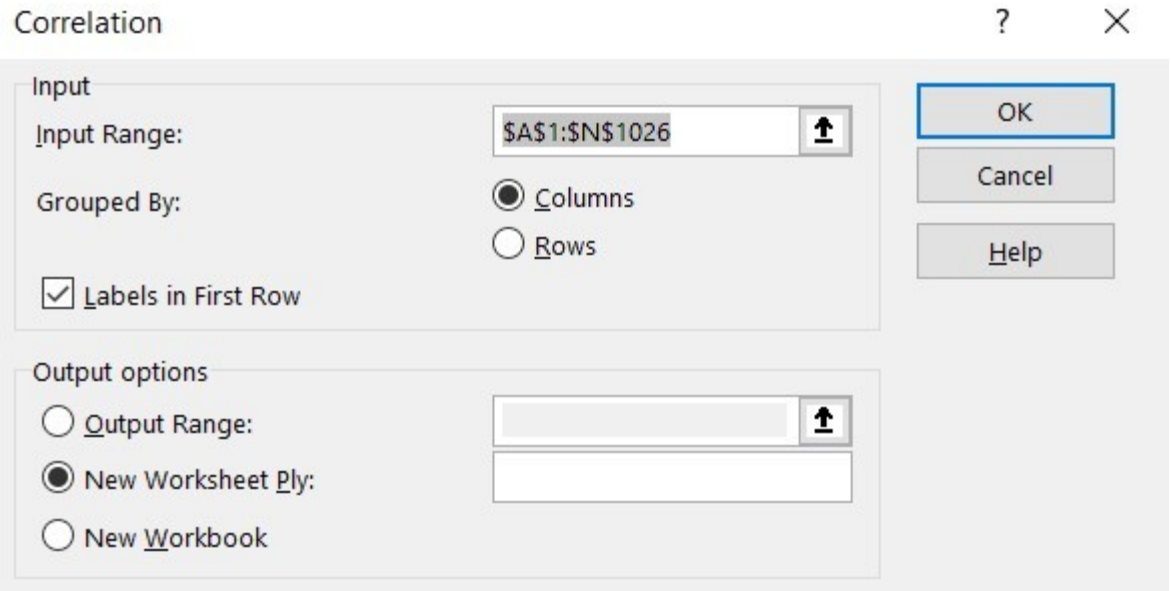
Step – 1 : Go to 'Data tab' and select 'Data analysis' from 'Analysis' section.



Step – 2 : Select 'correlation' and click OK.



Step – 3 : Select raw data in 'input range' section, select 'labels in the first row' and select 'New Worksheet Ply' and click OK.



Step – 4 : Generated result.

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
age	1													
sex	-0.10324	1												
cp	-0.07197	-0.04112	1											
trestbps	0.271121	-0.07897	0.038177	1										
chol	0.219823	-0.19826	-0.08164	0.127977	1									
fbs	0.121243	0.0272	0.079294	0.181767	0.026917	1								
restecg	-0.1327	-0.05512	0.043581	-0.12379	-0.14741	-0.10405	1							
thalach	-0.39023	-0.04937	0.306839	-0.03926	-0.02177	-0.00887	0.048411	1						
exang	0.088163	0.139157	-0.40151	0.061197	0.067382	0.049261	-0.06561	-0.38028	1					
oldpeak	0.208137	0.084687	-0.17473	0.187434	0.06488	0.010859	-0.05011	-0.3498	0.310844	1				
slope	-0.16911	-0.02667	0.131633	-0.12045	-0.01425	-0.0619	0.086086	0.395308	-0.26734	-0.57519	1			
ca	0.271551	0.111729	-0.17621	0.104554	0.074259	0.137156	-0.07807	-0.20789	0.107849	0.221816	-0.07344	1		
thal	0.072297	0.198424	-0.16334	0.059276	0.100244	-0.04218	-0.0205	-0.09807	0.197201	0.202672	-0.09409	0.149014	1	
target	-0.22932	-0.2795	0.434854	-0.13877	-0.09997	-0.04116	0.134468	0.422895	-0.43803	-0.43844	0.345512	-0.38209	-0.33784	1

SUMMARY

By applying the Karl Pearson's correlation method inbuilt in Excel we are able to see how the data is correlated to each other with the data analysis tool pack with the step-by-step procedure demonstrated above.

And in the data of the heart patients under target where 0 represents to no and 1 represents to yes which means what patient is a heart patient and who is not. With the correlation represented in the above image we can see how a variable is related to one another and provide proper values.

Here we're going to apply Karl Pearson's correlation method on few data sets such as following:-

The data below is a 2 variable distribution and numeric in nature.

Actually due to vast distribution we took sample data of height and weight of 30 students out of the population data of 88 students.

To perform correlation on the whole distribution, there are certain steps to be followed using data analysis tool pack using following steps

ASSIGNMENT – 7

Using Functions.

=RAND(), =RANDBETWEEN(MIN,MAX), =RANDARRAY([ROWS], [COLUMN],[MIN],[MAX],[INTEGER]).

Using ***=RAND()*** *we can get any random number generated by Excel such as if I apply this function I got a random number of* ***0.667278.***

Using ***=RANDBETWEEN(MIN,MAX)*** *we can get random number generated by Excel between minimum value and maximum value. Such as, if I apply this function with minimum value of* ***1*** *and maximum value of* ***50*** *I get a value of* ***18****.*

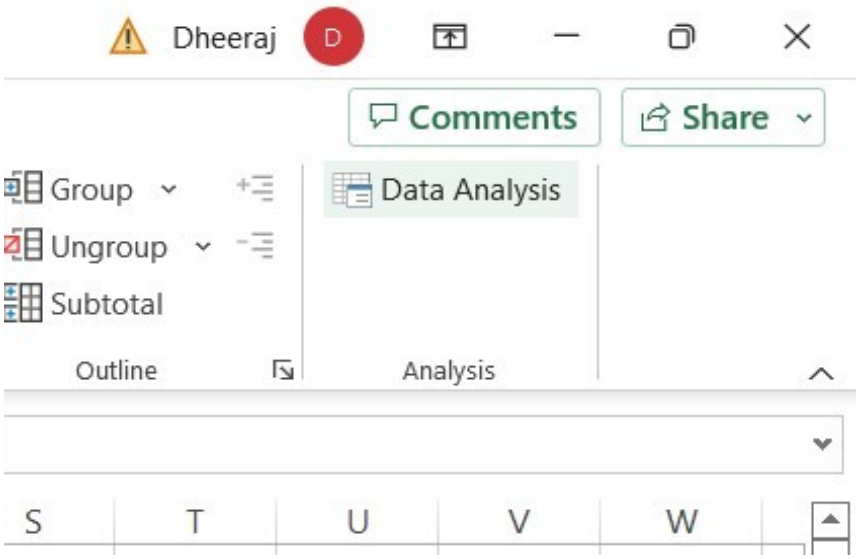
Using ***=RANDARRAY([ROWS],[COLUMN],[MIN],[MAX],[INTEGER])*** *we can get random number generated by Excel by initialising rows, column, minimum value, maximum value and integer value. Such as, if I apply this function with rows as* ***5****, column as* ***5****, minimum value as* ***1*** *and maximum value as* ***50*** *then I get the following table.*

47.04398	46.61656	23.95147	26.25522	9.742556
4.770054	41.72804	40.08385	37.08479	24.074
19.74278	25.90737	3.957669	33.88783	21.32983
28.12262	11.4302	39.81056	32.89275	46.89584
27.82627	46.97626	33.12953	30.02536	30.55776

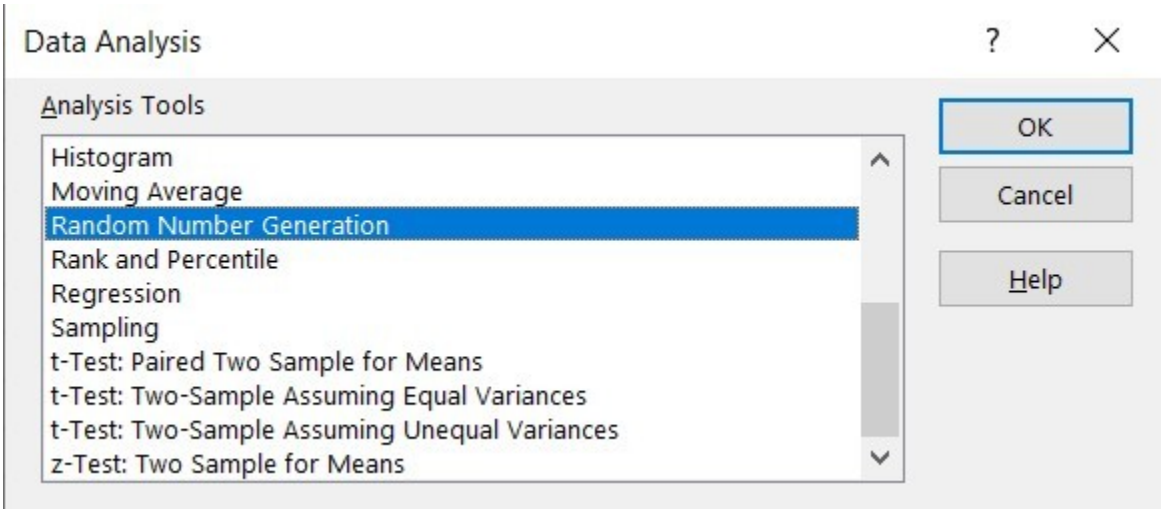
Using Data Analysis Toolpack.

By Normal distribution.

Step – 1 : Go to 'Data tab' and select 'Data analysis' from 'Analysis' section.

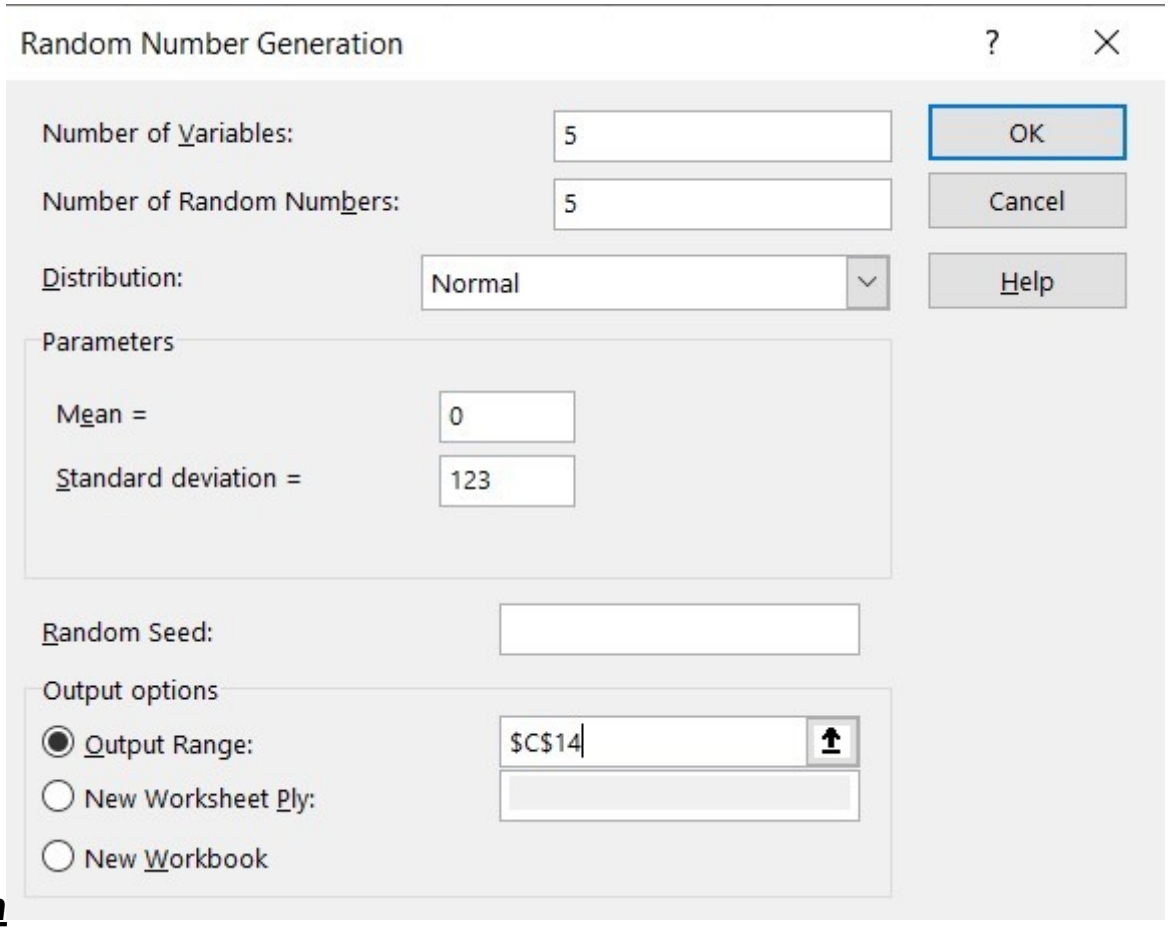


Step – 2 : Select 'Random Number Generation' option and click OK.



302.3161	-93.3744	60.41325	-83.3259	180.4305
41.35307	-59.5867	-49.8714	-110.927	172.2038
-57.2903	43.52023	24.25826	-101.368	-149.818
-126.872	152.2575	-114.478	-205.225	-228.003
-7.79132	144.9189	-105.523	197.4322	-89.6515

Step – 3 : Enter values

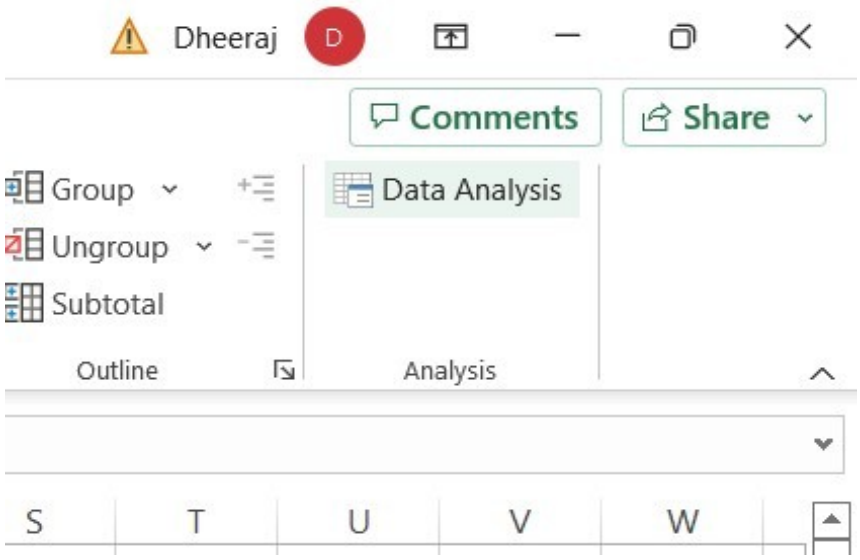


as m
entioned in picture below.

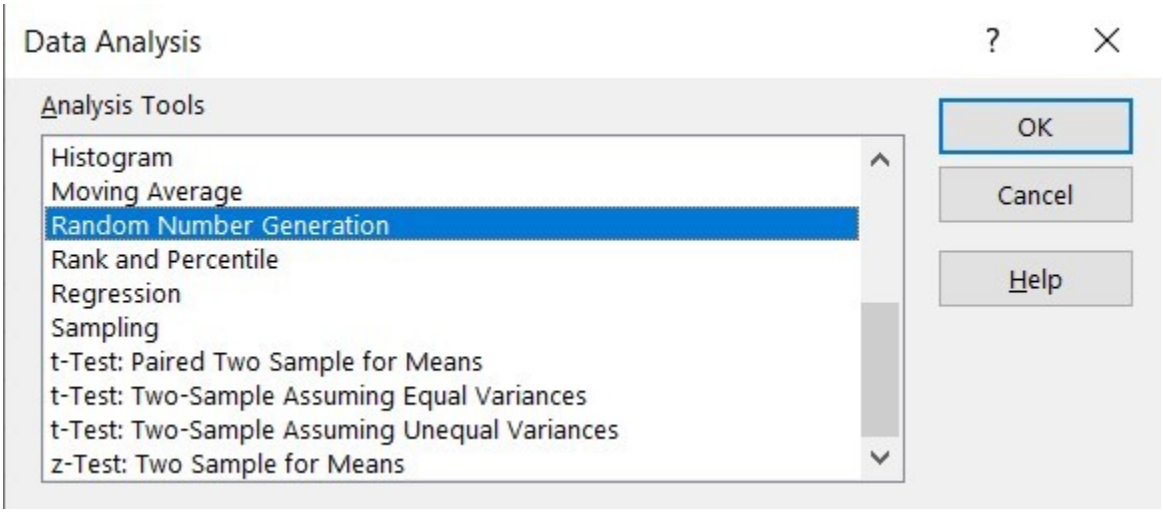
Step – 4 : Generated Result.

For Binomial Distribution.

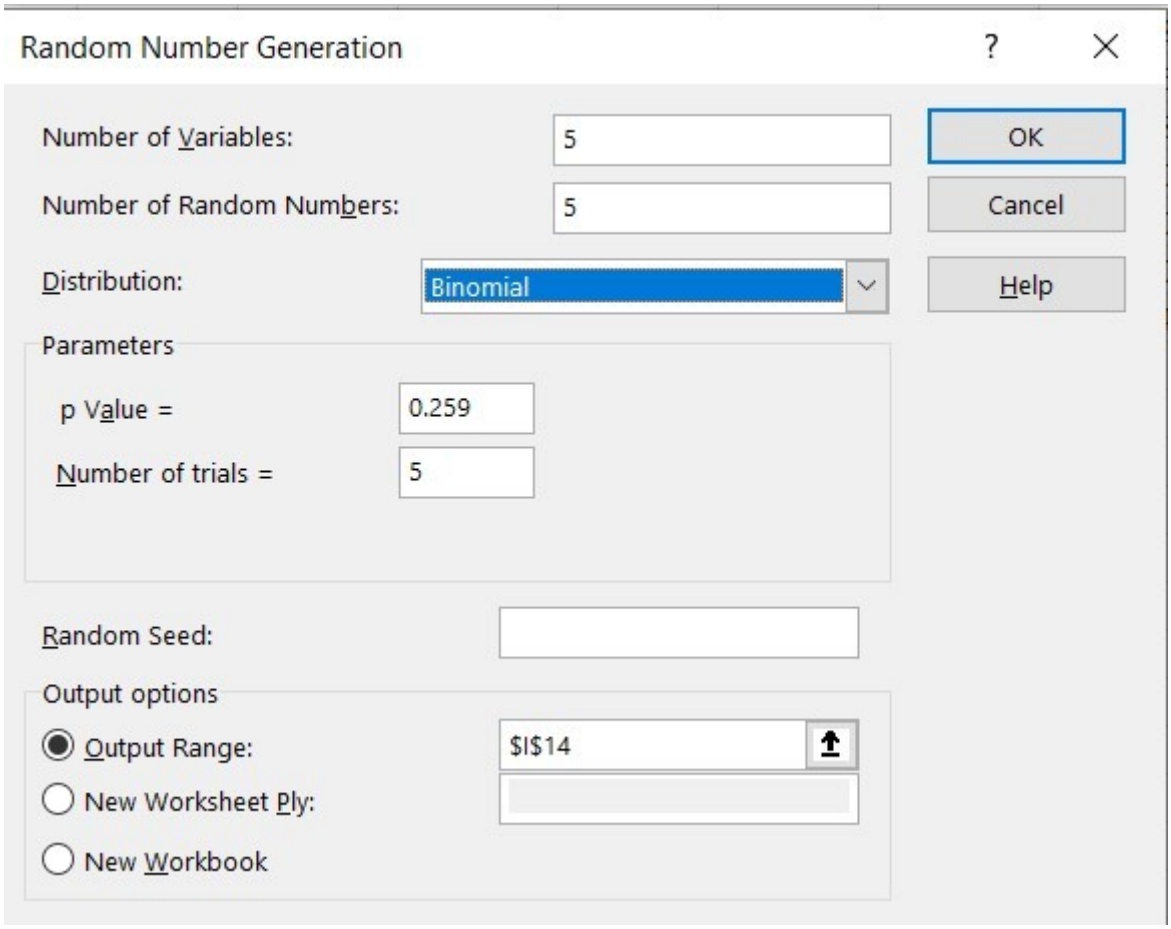
Step – 1 : Go to 'Data tab' and select 'Data analysis' from 'Analysis' section.



***Step – 2* : Select 'Random Number Generation' option and click OK.**



3	3	0	1	3	
1	1	1	0	1	
1	2	1	1	2	***Step – 3 : Enter values as mentioned in picture below.***
2	2	2	1	1	
0	2	0	1	0	


Random Number Generation
?
Number of Variables:
5
OK
Number of Random Numbers:
5
Cancel
Distribution:
Binomial
Help
Parameters
p Value =
0.259
Number of trials =
5
Random Seed:
Output options
Output Range:
I14
New Worksheet Ply:
New Workbook